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Units & Measurements

JEE Main Physics · Previous Year Questions with Answer Key

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MCQ (Single Correct Answer) 256 questions

Q1. JEE Main 2026 (Online) 8th April Evening Shift

A new unit (α) of length is chosen such that it is equal to the speed of light in vacuum. What is the distance between Venus and Earth in terms of α units if light takes 6 min. 40 s to cover this distance?

Open solution

(A) 200α (B) 400α (C) 300α (D) 500α

Q2. JEE Main 2026 (Online) 8th April Evening Shift

Consider the equation $H = \frac{x^p \epsilon^q E^r}{t^s}$

Where $H =$ magnetic field intensity; $E =$ electric field, $\epsilon =$ permittivity, $x =$ distance, $t =$ time The values of p, q, r and s respectively are :

Open solution

(A) $1, 1, 1, 1$ (B) $-1, 1, 2, 1$ (C) $1, -1, -2, 1$ (D) $-1, -2, -2, 1$

Q3. JEE Main 2026 (Online) 6th April Evening Shift

The percentage error in the calculated volume of a sphere, if there is 2% error in its diameter measurement, is ____ .

Open solution

(A) 1 (B) 2 (C) 6 (D) 8

Q4. JEE Main 2026 (Online) 6th April Evening Shift

Match List - I with List - II.

List - I		List - II	
A.	Boltzmann constant	I.	$\left[\mathrm{M}^{-1}\,\mathrm{L}^{3}\,\mathrm{T}^{-2}\right]$
B.	Stefan's constant	II.	$\left[\mathrm{M}\,\mathrm{L}^{2}\,\mathrm{T}^{-1}\right]$
C.	Planck's constant	III.	$\left[\mathrm{ML}^{2}\,\mathrm{T}^{-2}\,\mathrm{K}^{-1}\right]$
D.	Gravitational constant	IV.	$\left[\mathrm{M}\,\mathrm{L}^{0}\,\mathrm{T}^{-3}\,\mathrm{K}^{-4}\right]$

Open solution

Choose the correct answer from the options given below :

(A) A-I, B-II, C-III, D-IV (B) A-IV, B-III, C-II, D-I

(C) A-III, B-IV, C-II, D-I (D) A-II, B-I, C-IV, D-III

Q5. JEE Main 2026 (Online) 6th April Morning Shift

The density ρ of a uniform cylinder is determined by measuring its mass m, length l and diameter d. The measured values of m, l and d are 97.42 ± 0.02 g, 8.35 ± 0.05 mm and 20.20 ± 0.02 mm, respectively. Calculated percentage fractional error in ρ is ____ .

Open solution

(A) 0.63% (B) 0.82% (C) 0.72% (D) 0.25%

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Q6. JEE Main 2026 (Online) 6th April Morning Shift

Open solution

The potential energy of a particle changes with distance x from a fixed origin as $V = \frac{A\sqrt{x}}{x+B}$, where A and B are constants with appropriate dimensions. The dimensions of AB are ____

(A) $\left[\mathrm{M}^1\,\mathrm{L}^{5/2}\,\mathrm{T}^{-2}\right]$

(B) $\left[\mathrm{M}^{3/2}\,\mathrm{L}^{5/2}\,\mathrm{T}^{-2}\right]$

(C) $\left[\mathrm{M}^1\,\mathrm{L}^2\,\mathrm{T}^{-2}\right]$

(D) $\left[\mathrm{M}^1\,\mathrm{L}^{7/2}\,\mathrm{T}^{-2}\right]$

Q7. JEE Main 2026 (Online) 5th April Evening Shift

Open solution

Match List - I with List - II.

List - I		List - II	
A.	Meter (L)	I.	$\sqrt{\frac{hc}{G}}$
B.	Second (S)	II.	$\sqrt{\frac{Gh}{c^5}}$
C.	Kilogram (M)	III.	$\sqrt{\frac{K^2L^2c^3}{Gh}}$
D.	Kelvin (K)	IV.	$\sqrt{\frac{Gh}{c^3}}$

where h (Planck's constant), G (gravitational constant) and c (speed of light in vacuum) as fundamental units.

Choose the correct answer from the options given below :

(A) A-II, B-IV, C-I, D-III

(B) A-IV, B-II, C-I, D-III

(C) A-IV, B-I, C-II, D-III

(D) A-III, B-I, C-II, D-IV

Q8. JEE Main 2026 (Online) 5th April Evening Shift

Open solution

In an experiment to determine the resistance of a given wire using Ohm's law, the voltmeter and ammeter readings are noted as 10 V and 5 A , respectively. The least counts of voltmeter and ammeter are 500 mV and 200 mA , respectively. The estimated error in the resistance measurement is ____ Ω

(A) 0.25 (B) 2 (C) 2.5 (D) 0.18

Q9. JEE Main 2026 (Online) 5th April Morning Shift

Open solution

In a Vernier calipers, when both jaws touch each other, zero of the Vernier scale is shifted to the right of zero of the main scale and 7th Vernier division coincides with a main scale reading. If the value of 1 main scale division is 1 mm and there are 10 Vernier scale divisions, then the Vernier caliper has

(A) 0.07 cm negative zero error

(B) 0.7 cm negative zero error

(C) 0.07 cm positive zero error

(D) 0.7 cm positive zero error

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Q10. JEE Main 2026 (Online) 5th April Morning Shift

L, C and R represents physical quantities inductance, capacitance and resistance respectively. The dimensional formula $\mathrm{ML^2\,T^{-4}\,A^{-2}}$ corresponds to ____ .

(A) $\frac{R}{\sqrt{LC}}$ (B) $\frac{R}{LC}$ (C) $\frac{C}{\sqrt{LR}}$ (D) $\frac{1}{R}\sqrt{\frac{L}{C}}$

Open solution

Q11. JEE Main 2026 (Online) 4th April Evening Shift

Match the LIST-I with LIST-II

List - I		List - II	
A.	Planck's constant	I.	$\mathrm{ML^2\,T^{-2}}$
B.	Stopping potential	II.	$\mathrm{T^{-1}}$
C.	Work function	III.	$\mathrm{ML^2\,T^{-1}}$
D.	Threshold frequency	IV.	$\mathrm{ML^2\,T^{-3}\,A^{-1}}$

Choose the correct answer from the options given below:

(A) A-III, B-IV, C-I, D-II (B) A-I, B-II, C-III, D-IV

(C) A-IV, B-III, C-I, D-II (D) A-I, B-IV, C-III, D-II

Open solution

Q12. JEE Main 2026 (Online) 4th April Morning Shift

In a screw gauge when the circular scale is given five complete rotations it moves linearly by 2.5 mm . If the circular scale has 100 divisions, the least count of screw gauge is ____ mm.

(A) 1×10^{-2} (B) 1×10^{-3} (C) 5×10^{-2} (D) 5×10^{-3}

Open solution

Q13. JEE Main 2026 (Online) 2nd April Evening Shift

Dimensions of universal gravitational constant (G) in terms of Planck's constant (h), distance (L), mass (M) and time (T) are _____.

(A) $[hTLM^{-2}]$ (B) $[hT^{-1}LM^{-2}]$ (C) $[hTL^2M^{-2}]$ (D) $[h^{-1}T^{-1}LM^{-2}]$

Open solution

Q14. JEE Main 2026 (Online) 2nd April Evening Shift

In a screw gauge the zero of main scale reference line coincides with the fifth division of the circular scale when two studs are in contact. There are 100 divisions in circular scale and pitch of screw gauge is 0.1 mm. When diameter of a sphere is measured, the reading of main scale is 5 mm and 50th division of circular scale coincides with the reference line of main scale. The diameter of sphere is _____ mm.

(A) 5.045 (B) 5.055 (C) 5.450 (D) 5.550

Open solution

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Q15. JEE Main 2026 (Online) 2nd April Morning Shift

The dimensional formula of $\frac{1}{2}\epsilon_0 E^2$ (ϵ_0 = permittivity of vacuum and E = electric field) is $M^a L^b T^c$.

The value of $2a - b + c =$ _____.

(A) 0 (B) 1 (C) -1 (D) 2

Open solution

Q16. JEE Main 2026 (Online) 2nd April Morning Shift

The diameter of a wire measured by a screw gauge of least count 0.001 cm is 0.08 cm. The length measured by a scale of least count 0.1 cm is 150 cm. When a weight of 100 N is applied to the wire, the extension in length is 0.5 cm, measured by a micrometer of least count 0.001 cm. The error in the measured Young's modulus is $\alpha \times 10^9\ \mathrm{N/m^2}$. The value of α is _____.

(Ignore the contribution of the load to Young's modulus error calculation)

(A) 1.3 (B) 1.65 (C) 0.13 (D) 0.25

Open solution

Q17. JEE Main 2026 (Online) 28th January Evening Shift

The time period of a simple harmonic oscillator is $T = 2\pi\sqrt{\frac{k}{m}}$. Measured value of mass (m) of the object is 10 g with an accuracy of 10 mg and time for 50 oscillations of the spring is found to be 60 s using a watch of 2 s resolution. Percentage error in determination of spring constant (k) is _____%.

(A) 3.43 (B) 7.60 (C) 3.35 (D) 6.76

Open solution

Q18. JEE Main 2026 (Online) 28th January Evening Shift

The speed of a longitudinal wave in a metallic bar is 400 m/s. If the density and Young's modulus of the bar material are increased by 0.5% and 1%, respectively then the speed of the wave is changed approximately to _____ m/s.

(A) 398 (B) 402 (C) 401 (D) 399

Open solution

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Q19. JEE Main 2026 (Online) 28th January Evening Shift

Open solution

Match **List - I** with **List - II**.

List – I	List – II
A. Coefficient of viscosity	I. $[ML^{-1}T^{-2}]$
B. Surface tension	II. $[ML^{2}T^{-2}]$
C. Pressure	III. $[ML^{0}T^{-2}]$
D. Surface energy	IV. $[ML^{-1}T^{-1}]$

Choose the **correct** answer from the options given below :

(A) A-I, B-II, C-IV, D-III

(B) A-IV, B-III, C-I, D-II

(C) A-IV, B-I, C-II, D-III

(D) A-I, B-III, C-II, D-IV

Q20. JEE Main 2026 (Online) 28th January Evening Shift

Open solution

In an experiment, a set of readings are obtained as follows: 1.24 mm, 1.25 mm, 1.23 mm, 1.21 mm. The expected least count of the instrument used in recording these readings is _____ mm.

(A) 0.01

(B) 0.05

(C) 0.001

(D) 0.1

Q21. JEE Main 2026 (Online) 28th January Morning Shift

Open solution

When both jaws of vernier callipers touch each other, zero mark of the vernier scale is right to zero mark of main scale, 4th mark on vernier scale coincides with certain mark on the main scale. While measuring the length of a cylinder, observer observes 15 divisions on main scale and 5th division of vernier scale coincides with a main scale division. Measured length of cylinder is ____ mm.

(Least count of Vernier calliper $= 0.1\ \text{mm}$)

(A) 15.5

(B) 15.4

(C) 15.9

(D) 15.1

Q22. JEE Main 2026 (Online) 24th January Evening Shift

Open solution

In a vernier callipers, 50 vernier scale divisions are equal to 48 main scale divisions. If one main scale division $= 0.05\ \text{mm}$, then the least count of the vernier callipers is ____ mm.

(A) 0.002

(B) 0.02

(C) 0.05

(D) 0.005

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Q23. JEE Main 2026 (Online) 24th January Morning Shift

Open solution

Match the LIST-I with LIST-II

List-I		List-II	
A.	Magnetic induction	I.	$MLT^{-2}A^{-2}$
B.	Magnetic flux	II.	$ML^2T^{-2}A^{-2}$
C.	Magnetic permeability	III.	$ML^0T^{-2}A^{-1}$
D.	Self inductance	IV.	$ML^2T^{-2}A^{-1}$

Choose the correct answer from the options given below:

(A) A-III, B-IV, C-II, D-I
(B) A-I, B-III, C-IV, D-II
(C) A-IV, B-III, C-I, D-II
(D) A-III, B-IV, C-I, D-II

Q24. JEE Main 2026 (Online) 23rd January Evening Shift

Open solution

To compare EMF of two cells using potentiometer the balancing lengths obtained are 200 cm and 150 cm . The least count of scale is 1 cm . The percentage error in the ratio of EMFs is ____

(A) 1.75
(B) 1.55
(C) 1.17
(D) 1.65

Q25. JEE Main 2026 (Online) 23rd January Morning Shift

Open solution

In a screw gauge, the zero of the circular scale lies 3 divisions above the horizontal pitch line when their metallic studs are brought in contact. Using this instrument thickness of a sheet is measured. If pitch scale reading is 1 mm and the circular scale reading is 51 then the correct thickness of the sheet is ____ mm.

[Assume least count is 0.01 mm]

(A) 1.54
(B) 1.50
(C) 1.51
(D) 1.48

Q26. JEE Main 2026 (Online) 23rd January Morning Shift

Open solution

Four persons measure the length of a rod as $20.00\ \text{cm}, 19.75\ \text{cm}, 17.01\ \text{cm}$ and 18.25 cm . The relative error in the measurement of average length of the rod is :

(A) 0.18
(B) 0.24
(C) 0.06
(D) 0.08

Q27. JEE Main 2026 (Online) 22nd January Evening Shift

Open solution

If ϵ, E and t represent the free space permittivity, electric field and time respectively, then the unit of $\frac{\epsilon E}{t}$ will be :

(A) Am
(B) A/m
(C) A/m^2
(D) Am^2

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Q28. JEE Main 2026 (Online) 22nd January Morning Shift

Open solution

Match the LIST-I with LIST-II

List-I		List-II	
A.	Spring constant	I.	$ML^2T^{-2}K^{-1}$
B.	Thermal conductivity	II.	ML^0T^{-2}
C.	Boltzmann constant	III.	$ML^2T^{-3}A^{-2}$
D.	Inductive reactance	IV.	$MLT^{-3}K^{-1}$

Choose the correct answer from the options given below:

(A) A-III, B-II, C-IV, D-I

(B) A-I, B-IV, C-II, D-III

(C) A-II, B-I, C-IV, D-III

(D) A-II, B-IV, C-I, D-III

Q29. JEE Main 2026 (Online) 21st January Evening Shift

Open solution

A spherical body of radius r and density σ falls freely through a viscous liquid having density ρ and viscosity η and attains a terminal velocity v_0. Estimated maximum error in the quantity η is : (Ignore errors associated with σ, ρ and g, gravitational acceleration)

(A) $2\left[\frac{\Delta r}{r}-\frac{\Delta v_0}{v_0}\right]$

(B) $2\left[\frac{\Delta r}{r}+\frac{\Delta v_0}{v_0}\right]$

(C) $\frac{2\Delta r}{r}+\frac{\Delta v_0}{v_0}$

(D) $2\frac{\Delta r}{r}-\frac{\Delta v_0}{v_0}$

Q30. JEE Main 2026 (Online) 21st January Evening Shift

Open solution

Keeping the significant figures in view, the sum of the physical quantities 52.01 m, 153.2 m and 0.123 m is :

(A) 205.3 m

(B) 205 m

(C) 205.333 m

(D) 205.33 m

Q31. JEE Main 2026 (Online) 21st January Morning Shift

Open solution

In an experiment the values of two spring constants were measured as $k_1=(10\pm0.2)\mathrm{N/m}$ and $k_2=(20\pm0.3)\mathrm{N/m}$. If these springs are connected in parallel, then the percentage error in equivalent spring constant is :

(A) 1.33%

(B) 2.67%

(C) 1.67%

(D) 2.33%

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Q32. JEE Main 2026 (Online) 21st January Morning Shift

Open solution

Consider a modified Bernoulli equation.

$\left(\mathrm{P}+\frac{A}{Bt^2}\right)+\rho g(h+Bt)+\frac{1}{2}\rho V^2=$ constant

If t has the dimension of time then the dimensions of A and B are ____ , ____ respectively.

(A) $\left[\mathrm{ML}^0\,\mathrm{T}^{-1}\right]$ and $\left[\mathrm{M}^0\mathrm{LT}\right]$

(B) $\left[\mathrm{ML}^0\,\mathrm{T}^{-2}\right]$ and $\left[\mathrm{M}^0\mathrm{LT}^{-1}\right]$

(C) $\left[\mathrm{ML}^0\,\mathrm{T}^{-2}\right]$ and $\left[\mathrm{M}^0\mathrm{LT}^{-2}\right]$

(D) $\left[\mathrm{ML}^0\,\mathrm{T}^{-1}\right]$ and $\left[\mathrm{M}^0\mathrm{LT}^{-1}\right]$

Q33. JEE Main 2025 (Online) 8th April Evening Shift

Open solution

A quantity Q is formulated as $X^{-2}Y^{+\frac{3}{2}}Z^{-\frac{2}{5}}$. X, Y, and Z are independent parameters which have fractional errors of 0.1, 0.2, and 0.5, respectively in measurement. The maximum fractional error of Q is

(A) 0.6 (B) 0.8 (C) 0.7 (D) 0.1

Q34. JEE Main 2025 (Online) 7th April Evening Shift

Open solution

The dimension of $\sqrt{\frac{\mu_0}{\epsilon_0}}$ is equal to that of:

(μ_0 = Vacuum permeability and ϵ_0 = Vacuum permittivity)

(A) Voltage (B) Inductance (C) Resistance (D) Capacitance

Q35. JEE Main 2025 (Online) 7th April Evening Shift

Open solution

Match **List - I** with **List - II**.

List - I	List - II
(A) Mass density	(I) $[ML^2T^{-3}]$
(B) Impulse	(II) $[MLT^{-1}]$
(C) Power	(III) $[ML^2T^0]$
(D) Moment of inertia	(IV) $[ML^{-3}T^0]$

Choose the **correct** answer from the options given below :

(A) (A)-(IV), (B)-(II), (C)-(III), (D)-(I)

(B) (A)-(II), (B)-(III), (C)-(IV), (D)-(I)

(C) (A)-(IV), (B)-(III), (C)-(I), (D)-(II)

(D) (A)-(IV), (B)-(II), (C)-(I), (D)-(III)

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Q36. JEE Main 2025 (Online) 4th April Evening Shift

In an electromagnetic system, a quantity defined as the ratio of electric dipole moment and magnetic dipole moment has dimension of $\left[\mathrm{M}^{\mathrm{P}} \mathrm{L}^{\mathrm{Q}} \mathrm{T}^{R} A^{\mathrm{S}}\right]$. The value of P and Q are :

(A) $-1, 0$ (B) $0, -1$ (C) $-1, 1$ (D) $1, -1$

Open solution

Q37. JEE Main 2025 (Online) 4th April Evening Shift

For the determination of refractive index of glass slab, a travelling microscope is used whose main scale contains 300 equal divisions equals to 15 cm . The vernier scale attached to the microscope has 25 divisions equals to 24 divisions of main scale. The least count (LC) of the travelling microscope is (in cm) :

(A) 0.002 (B) 0.0025 (C) 0.0005 (D) 0.001

Open solution

Q38. JEE Main 2025 (Online) 4th April Morning Shift

In an electromagnetic system, the quantity representing the ratio of electric flux and magnetic flux has dimension of $M^{P} L^{Q} T^{R} A^{S}$, where value of ' Q ' and ' R ' are

(A) $(3, -5)$ (B) $(-2, 1)$ (C) $(-2, 2)$ (D) $(1, -1)$

Open solution

Q39. JEE Main 2025 (Online) 3rd April Evening Shift

Match the LIST-I with LIST-II

LIST-I		LIST-II	
A.	Boltzmann constant	I	$\mathrm{ML}^{2} \mathrm{T}^{-1}$
B	Coefficient of viscosity	II	$\mathrm{MLT}^{-3} \mathrm{K}^{-1}$
C	Planck's constant	III	$\mathrm{ML}^{2} \mathrm{T}^{-2} \mathrm{K}^{-1}$
D	Thermal conductivity	IV	$\mathrm{ML}^{-1} \mathrm{T}^{-1}$

Choose the correct answer from the options given below:

(A) A - III, B - IV, C - I, D - II (B) A - III, B - IV, C - II, D - I

(C) A - III, B - II, C - I, D - IV (D) A - II, B - III, C - IV, D - I

Open solution

Q40. JEE Main 2025 (Online) 3rd April Morning Shift

A person measures mass of 3 different particles as $435.42\ \mathrm{g}, 226.3\ \mathrm{g}$ and 0.125 g . According to the rules for arithmetic operations with significant figures, the addition of the masses of 3 particles will be.

(A) 661.845 g (B) 661.84 g (C) 662 g (D) 661.8 g

Open solution

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Q41. JEE Main 2025 (Online) 3rd April Morning Shift

Open solution

Match the LIST-I with LIST-II

List - I		List - II	
A.	Gravitational constant	I.	$\left[\mathrm{LT}^{-2}\right]$
B.	Gravitational potential energy	II.	$\left[\mathrm{L}^{2}\,\mathrm{T}^{-2}\right]$
C.	Gravitational potential	III.	$\left[\mathrm{ML}^{2}\,\mathrm{T}^{-2}\right]$
D.	Acceleration due to gravity	IV.	$\left[\mathrm{M}^{-1}\,\mathrm{L}^{3}\,\mathrm{T}^{-2}\right]$

Choose the correct answer from the options given below:

(A) A-IV, B-III, C-II, D-I

(B) A-II, B-IV, C-III, D-I

(C) A-I, B-III, C-IV, D-II

(D) A-III, B-II, C-I, D-IV

Q42. JEE Main 2025 (Online) 2nd April Evening Shift

Open solution

If μ_0 and ϵ_0 are the permeability and permittivity of free space, respectively, then the dimension of $\left(\frac{1}{\mu_0\epsilon_0}\right)$ is :

(A) T^2/L

(B) $\mathrm{L}^2/\mathrm{T}^2$

(C) $\mathrm{T}^2/\mathrm{L}^2$

(D) L/T^2

Q43. JEE Main 2025 (Online) 2nd April Evening Shift

Open solution

Given a charge q , current I and permeability of vacuum μ_o. Which of the following quantity has the dimension of momentum ?

(A) qI/μ_o

(B) $q^2\mu_o I$

(C) $q\mu_0/I$

(D) $\mathrm{q}\mu_o\mathrm{I}$

Q44. JEE Main 2025 (Online) 2nd April Morning Shift

Open solution

Match List I with List II.

	List - I		List - II
(A)	Coefficient of viscosity	(I)	$\left[\mathrm{ML}^{0}\,\mathrm{T}^{-3}\right]$
(B)	Intensity of wave	(II)	$\left[\mathrm{ML}^{-2}\,\mathrm{T}^{-2}\right]$
(C)	Pressure gradient	(III)	$\left[\mathrm{M}^{-1}\mathrm{LT}^{2}\right]$
(D)	Compressibility	(IV)	$\left[\mathrm{ML}^{-1}\,\mathrm{T}^{-1}\right]$

Choose the correct answer from the options given below:

(A) (A)-(II), (B)-(III), (C)-(IV), (D)-(I)

(B) (A)-(IV), (B)-(II), (C)-(I), (D)-(III)

(C) (A)-(IV), (B)-(I), (C)-(II), (D)-(III)

(D) (A)-(I), (B)-(IV), (C)-(III), (D)-(II)

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Q45. JEE Main 2025 (Online) 2nd April Morning Shift

The equation for real gas is given by $\left(P+\frac{a}{V^2}\right)(V-b)=RT$, where P, V, T and R are the pressure, volume, temperature and gas constant, respectively. The dimension of ab^{-2} is equivalent to that of :

Open solution

(A) Compressibility
(B) Planck's constant
(C) Energy density
(D) Strain

Q46. JEE Main 2025 (Online) 29th January Evening Shift

Match **List - I** with **List - II**.

List - I	List - II
(A) Young's Modulus	(I) $M L^{-1} T^{-1}$
(B) Torque	(II) $M L^{-1} T^{-2}$
(C) Coefficient of Viscosity	(III) $M^{-1} L^{3} T^{-2}$
(D) Gravitational Constant	(IV) $M L^{2} T^{-2}$

Open solution

Choose the **correct** answer from the options given below :

(A) (A)-(I), (B)-(III), (C)-(II), (D)-(IV)
(B) (A)-(IV), (B)-(II), (C)-(III), (D)-(I)
(C) (A)-(II), (B)-(I), (C)-(IV), (D)-(III)
(D) (A)-(II), (B)-(IV), (C)-(I), (D)-(III)

Q47. JEE Main 2025 (Online) 29th January Morning Shift

The pair of physical quantities not having the same dimensions is :

Open solution

(A) Angular momentum and Planck's constant
(B) Torque and energy
(C) Surface tension and impulse
(D) Pressure and Young's modulus

Q48. JEE Main 2025 (Online) 29th January Morning Shift

The expression given below shows the variation of velocity (v) with time (t),

$v=At^2+\frac{Bt}{C+t}$.

The dimension of ABC is :

Open solution

(A) $[M^0L^1T^{-2}]$
(B) $[M^0L^2T^{-3}]$
(C) $[M^0L^2T^{-2}]$
(D) $[M^0L^1T^{-3}]$

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Q49. JEE Main 2025 (Online) 28th January Evening Shift

Open solution

Match List - I with List - II.

List - I	List - II
(A) Angular Impulse	(I) $M^0 L^2 T^{-2}$
(B) Latent Heat	(II) $M L^2 T^{-3} A^{-1}$
(C) Electrical resistivity	(III) $M L^2 T^{-1}$
(D) Electromotive force	(IV) $M L^3 T^{-3} A^{-2}$

Choose the correct answer from the options given below:

(A) (A)-(II), (B)-(I), (C)-(IV), (D)-(III)

(B) (A)-(I), (B)-(III), (C)-(IV), (D)-(II)

(C) (A)-(III), (B)-(I), (C)-(IV), (D)-(II)

(D) (A)-(III), (B)-(I), (C)-(II), (D)-(IV)

Q50. JEE Main 2025 (Online) 24th January Evening Shift

Open solution

The energy E and momentum p of a moving body of mass m are related by some equation. Given that c represents the speed of light, identify the correct equation

(A) $\mathrm{E}^2 = \mathrm{pc}^2 + \mathrm{m}^2\mathrm{c}^4$

(B) $\mathrm{E}^2 = \mathrm{pc}^2 + \mathrm{m}^2\mathrm{c}^2$

(C) $\mathrm{E}^2 = \mathrm{p}^2\mathrm{c}^2 + \mathrm{m}^2\mathrm{c}^4$

(D) $\mathrm{E}^2 = \mathrm{p}^2\mathrm{c}^2 + \mathrm{m}^2\mathrm{c}^2$

Q51. JEE Main 2025 (Online) 24th January Morning Shift

Open solution

For an experimental expression $y = \frac{32.3 \times 1125}{27.4}$, where all the digits are significant. Then to report the value of y we should write

(A) $y = 1326.186$

(B) $y = 1326.2$

(C) $y = 1326.19$

(D) $y = 1330$

Q52. JEE Main 2025 (Online) 23rd January Evening Shift

Open solution

The energy of a system is given as $\mathrm{E(t)} = \alpha^3 \mathrm{e}^{-\beta t}$, where t is the time and $\beta = 0.3\,\mathrm{s}^{-1}$. The errors in the measurement of α and t are 1.2% and 1.6%, respectively. At $t = 5\,\mathrm{s}$, maximum percentage error in the energy is :

(A) 6%

(B) 11.6%

(C) 4%

(D) 8.4%

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Q53. JEE Main 2025 (Online) 23rd January Evening Shift

Open solution

Match List - I with List - II

	List - I		List - II
(A)	Permeability of free space	(I)	$\left[\mathrm{ML^2\,T^{-2}}\right]$
(B)	Magnetic field	(II)	$\left[\mathrm{MT^{-2}\,A^{-1}}\right]$
(C)	Magnetic moment	(III)	$\left[\mathrm{MLT^{-2}\,A^{-2}}\right]$
(D)	Torsional constant	(IV)	$\left[\mathrm{L^2\,A}\right]$

Choose the correct answer from the options given below :

(A) (A)-(III), (B)-(II), (C)-(IV), (D)-(I)

(B) $(\mathrm{A})-(\mathrm{I}),(\mathrm{B})-(\mathrm{IV}),(\mathrm{C})-(\mathrm{II}),(\mathrm{D})-(\mathrm{III})$

(C) $(\mathrm{A})-(\mathrm{II}),(\mathrm{B})-(\mathrm{I}),(\mathrm{C})-(\mathrm{III}),(\mathrm{D})-(\mathrm{IV})$

(D) (A)-(IV), (B)-(III), (C)-(I), (D)-(II)

Q54. JEE Main 2025 (Online) 23rd January Morning Shift

Open solution

The position of a particle moving on x-axis is given by $x(t) = A\sin t + B\cos^2 t + Ct^2 + D$, where t is time. The dimension of $\frac{ABC}{D}$ is

(A) L **(B)** $\mathrm{L^3\,T^{-2}}$ **(C)** $\mathrm{L^2\,T^{-2}}$ **(D)** $\mathrm{L^2}$

Q55. JEE Main 2025 (Online) 23rd January Morning Shift

Open solution

The electric flux is $\phi = \alpha\sigma + \beta\lambda$ where λ and σ are linear and surface charge density, respectively. $\left(\frac{\alpha}{\beta}\right)$ represents

(A) displacement **(B)** charge **(C)** electric field **(D)** area

Q56. JEE Main 2025 (Online) 22nd January Evening Shift

Open solution

Which one of the following is the correct dimensional formula for the capacitance in F ? $\mathrm{M}, \mathrm{L}, \mathrm{T}$ and C stand for unit of mass, length, time and charge respectively.

(A) $[\mathrm{F}] = \left[\mathrm{CM^{-1}\,L^{-2}\,T^2}\right]$

(B) $[\mathrm{F}] = \left[\mathrm{C^2M^{-2}\,L^2\,T^2}\right]$

(C) $[\mathrm{F}] = \left[\mathrm{C^2M^{-1}\,L^{-2}\,T^2}\right]$

(D) $[\mathrm{F}] = \left[\mathrm{CM^{-2}\,L^{-2}\,T^{-2}}\right]$

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Q57. JEE Main 2025 (Online) 22nd January Evening Shift

The maximum percentage error in the measurement of density of a wire is

[Given, mass of wire $= (0.60 \pm 0.003)\text{g}$

radius of wire $= (0.50 \pm 0.01)\text{cm}$

length of wire $= (10.00 \pm 0.05)\text{cm}$]

(A) 7 (B) 8 (C) 5 (D) 4

Open solution

Q58. JEE Main 2025 (Online) 22nd January Morning Shift

If B is magnetic field and μ_0 is permeability of free space, then the dimensions of (B/μ_0) is

(A) $\text{MT}^{-2}\,\text{A}^{-1}$ (B) $\text{L}^{-1}\,\text{A}$

(C) $\text{ML}^2\,\text{T}^{-2}\,\text{A}^{-1}$ (D) $\text{LT}^{-2}\,\text{A}^{-1}$

Open solution

Q59. JEE Main 2025 (Online) 22nd January Morning Shift

Given below are two statements :

Statement I: In a vernier callipers, one vernier scale division is always smaller than one main scale division.

Statement II : The vernier constant is given by one main scale division multiplied by the number of vernier scale divisions.

In the light of the above statements, choose the correct answer from the options given below.

(A) Both Statement I and Statement II are false

(B) Statement I is true but Statement II is false

(C) Both Statement I and Statement II are true

(D) Statement I is false but Statement II is true

Open solution

Q60. JEE Main 2024 (Online) 9th April Evening Shift

The de-Broglie wavelength associated with a particle of mass m and energy E is $h/\sqrt{2mE}$. The dimensional formula for Planck's constant is :

(A) $\left[\text{M}^2\,\text{L}^2\,\text{T}^{-2}\right]$ (B) $\left[\text{ML}^{-1}\,\text{T}^{-2}\right]$

(C) $\left[\text{ML}^2\,\text{T}^{-1}\right]$ (D) $\left[\text{MLT}^{-2}\right]$

Open solution

Q61. JEE Main 2024 (Online) 9th April Morning Shift

The dimensional formula of latent heat is :

(A) $\left[\text{M}^0\text{LT}^{-2}\right]$ (B) $\left[\text{MLT}^{-2}\right]$

(C) $\left[\text{M}^0\,\text{L}^2\,\text{T}^{-2}\right]$ (D) $\left[\text{ML}^2\,\text{T}^{-2}\right]$

Open solution

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Q62. JEE Main 2024 (Online) 9th April Morning Shift

One main scale division of a vernier caliper is equal to m units. If n^{th} division of main scale coincides with $(n+1)^{\text{th}}$ division of vernier scale, the least count of the vernier caliper is :

(A) $\frac{1}{(\mathrm{n}+1)}$
(B) $\frac{m}{(n+1)}$
(C) $\frac{n}{(n+1)}$
(D) $\frac{\mathrm{m}}{\mathrm{n}(\mathrm{n}+1)}$

Open solution

Q63. JEE Main 2024 (Online) 8th April Evening Shift

If ϵ_{o} is the permittivity of free space and E is the electric field, then $\epsilon_{\mathrm{o}}\mathrm{E}^2$ has the dimensions :

(A) $[\mathrm{ML}^{-1}\,\mathrm{T}^{-2}]$
(B) $[\mathrm{ML}^2\,\mathrm{T}^{-2}]$
(C) $[\mathrm{M}^0\mathrm{L}^{-2}\mathrm{TA}]$
(D) $[\mathrm{M}^{-1}\,\mathrm{L}^{-3}\,\mathrm{T}^4\,\mathrm{A}^2]$

Open solution

Q64. JEE Main 2024 (Online) 8th April Evening Shift

There are 100 divisions on the circular scale of a screw gauge of pitch $1\ \mathrm{mm}$. With no measuring quantity in between the jaws, the zero of the circular scale lies 5 divisions below the reference line. The diameter of a wire is then measured using this screw gauge. It is found that 4 linear scale divisions are clearly visible while 60 divisions on circular scale coincide with the reference line. The diameter of the wire is :

(A) 4.65 mm
(B) 4.60 mm
(C) 4.55 mm
(D) 3.35 mm

Open solution

Q65. JEE Main 2024 (Online) 8th April Evening Shift

Least count of a vernier caliper is $\frac{1}{20\,\mathrm{N}}\ \mathrm{cm}$. The value of one division on the main scale is $1\ \mathrm{mm}$. Then the number of divisions of main scale that coincide with N divisions of vernier scale is :

(A) $\left(\frac{2\,\mathrm{N}-1}{2\,\mathrm{N}}\right)$
(B) $\left(\frac{2\,\mathrm{N}-1}{20\,\mathrm{N}}\right)$
(C) $(2\,\mathrm{N}-1)$
(D) $\left(\frac{2\,\mathrm{N}-1}{2}\right)$

Open solution

Q66. JEE Main 2024 (Online) 8th April Morning Shift

In an expression $a \times 10^b$:

(A) a is order of magnitude for $b \leq 5$
(B) b is order of magnitude for $a \leq 5$
(C) b is order of magnitude for $a \geq 5$
(D) b is order of magnitude for $5 < a \leq 10$

Open solution

Q67. JEE Main 2024 (Online) 8th April Morning Shift

Young's modulus is determined by the equation given by $\mathrm{Y} = 49000\frac{\mathrm{m}}{\mathrm{l}}\frac{\mathrm{dyne}}{\mathrm{cm}^2}$ where M is the mass and l is the extension of wire used in the experiment. Now error in Young modules (Y) is estimated by taking data from $M-l$ plot in graph paper. The smallest scale divisions are $5\ \mathrm{g}$ and $0.02\ \mathrm{cm}$ along load axis and extension axis respectively. If the value of M and l are $500\ \mathrm{g}$ and $2\ \mathrm{cm}$ respectively then percentage error of Y is :

(A) 2%
(B) 0.02%
(C) 0.5%
(D) 0.2%

Open solution

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Q68. JEE Main 2024 (Online) 8th April Morning Shift

Open solution

The diameter of a sphere is measured using a vernier caliper whose 9 divisions of main scale are equal to 10 divisions of vernier scale. The shortest division on the main scale is equal to $1\ \text{mm}$. The main scale reading is $2\ \text{cm}$ and second division of vernier scale coincides with a division on main scale. If mass of the sphere is $8.635\ \text{g}$, the density of the sphere is:

(A) $2.2\ \text{g/cm}^3$ (B) $2.0\ \text{g/cm}^3$ (C) $1.7\ \text{g/cm}^3$ (D) $2.5\ \text{g/cm}^3$

Q69. JEE Main 2024 (Online) 6th April Evening Shift

Open solution

Given below are two statements:

Statement (I) : Dimensions of specific heat is $[\text{L}^2\ \text{T}^{-2}\ \text{K}^{-1}]$.

Statement (II) : Dimensions of gas constant is $[\text{ML}^2\ \text{T}^{-1}\ \text{K}^{-1}]$.

In the light of the above statements, choose the most appropriate answer from the options given below.

(A) Statement (I) is incorrect but statement (II) is correct

(B) Both statement (I) and statement (II) are incorrect

(C) Both statement (I) and statement (II) are correct

(D) Statement (I) is correct but statement (II) is incorrect

Q70. JEE Main 2024 (Online) 6th April Evening Shift

Open solution

In a vernier calliper, when both jaws touch each other, zero of the vernier scale shifts towards left and its 4th division coincides exactly with a certain division on main scale. If 50 vernier scale divisions equal to 49 main scale divisions and zero error in the instrument is $0.04\ \text{mm}$ then how many main scale divisions are there in $1\ \text{cm}$?

(A) 5 (B) 40 (C) 10 (D) 20

Q71. JEE Main 2024 (Online) 6th April Evening Shift

Open solution

In finding out refractive index of glass slab the following observations were made through travelling microscope 50 vernier scale division $= 49\ \text{MSD}; 20$ divisions on main scale in each cm

For mark on paper

$\text{MSR} = 8.45\ \text{cm}, \text{VC} = 26$

For mark on paper seen through slab

$\text{MSR} = 7.12\ \text{cm}, \text{VC} = 41$

For powder particle on the top surface of the glass slab

$\text{MSR} = 4.05\ \text{cm}, \text{VC} = 1$

(MSR = Main Scale Reading, VC = Vernier Coincidence)

Refractive index of the glass slab is :

(A) 1.52 (B) 1.35 (C) 1.24 (D) 1.42

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Q72. JEE Main 2024 (Online) 6th April Morning Shift

To find the spring constant (k) of a spring experimentally, a student commits 2% positive error in the measurement of time and 1% negative error in measurement of mass. The percentage error in determining value of k is :

Open solution

(A) 5% (B) 3% (C) 1% (D) 4%

Q73. JEE Main 2024 (Online) 6th April Morning Shift

Match List I with List II

	LIST I		LIST II
A.	Torque	I.	$[M^1L^1T^{-2}A^{-2}]$
B.	Magnetic field	II.	$[L^2A^1]$
C.	Magnetic moment	III.	$[M^1T^{-2}A^{-1}]$
D.	Permeability of free space	IV.	$[M^1L^2T^{-2}]$

Open solution

Choose the correct answer from the options given below:

(A) A-I, B-III, C-II, D-IV (B) A-IV, B-II, C-III, D-I

(C) A-III, B-I, C-II, D-IV (D) A-IV, B-III, C-II, D-I

Q74. JEE Main 2024 (Online) 6th April Morning Shift

While measuring diameter of wire using screw gauge the following readings were noted. Main scale reading is $1\ \text{mm}$ and circular scale reading is equal to 42 divisions. Pitch of screw gauge is $1\ \text{mm}$ and it has 100 divisions on circular scale. The diameter of the wire is $\frac{x}{50}\ \text{mm}$. The value of x is :

Open solution

(A) 42 (B) 71 (C) 21 (D) 142

Q75. JEE Main 2024 (Online) 5th April Evening Shift

What is the dimensional formula of ab^{-1} in the equation $\left(\mathrm{P}+\frac{\mathrm{a}}{\mathrm{V}^2}\right)(\mathrm{V}-\mathrm{b})=\mathrm{RT}$, where letters have their usual meaning.

Open solution

(A) $[\mathrm{M}^6\,\mathrm{L}^7\,\mathrm{T}^4]$ (B) $[\mathrm{M}^{-1}\,\mathrm{L}^5\,\mathrm{T}^3]$

(C) $[\mathrm{M}^0\,\mathrm{L}^3\,\mathrm{T}^{-2}]$ (D) $[\mathrm{ML}^2\,\mathrm{T}^{-2}]$

Q76. JEE Main 2024 (Online) 5th April Evening Shift

A vernier callipers has 20 divisions on the vernier scale, which coincides with 19th division on the main scale. The least count of the instrument is $0.1\ \text{mm}$. One main scale division is equal to _____ mm.

Open solution

(A) 5 (B) 2 (C) 1 (D) 0.5

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Q77. JEE Main 2024 (Online) 5th April Morning Shift

If G be the gravitational constant and u be the energy density then which of the following quantity have the dimensions as that of the $\sqrt{\mathrm{uG}}$:

(A) Gravitational potential

(B) pressure gradient per unit mass

(C) Energy per unit mass

(D) Force per unit mass

Open solution

Q78. JEE Main 2024 (Online) 5th April Morning Shift

Time periods of oscillation of the same simple pendulum measured using four different measuring clocks were recorded as $4.62\ \mathrm{s}, 4.632\ \mathrm{s}, 4.6\ \mathrm{s}$ and $4.64\ \mathrm{s}$. The arithmetic mean of these readings in correct significant figure is :

(A) 5 s

(B) 4.6 s

(C) 4.62 s

(D) 4.623 s

Open solution

Q79. JEE Main 2024 (Online) 4th April Evening Shift

Applying the principle of homogeneity of dimensions, determine which one is correct, where T is time period, G is gravitational constant, M is mass, r is radius of orbit.

(A) $T^2 = \frac{4\pi^2 r}{GM^2}$

(B) $T^2 = 4\pi^2 r^3$

(C) $T^2 = \frac{4\pi^2 r^3}{GM}$

(D) $T^2 = \frac{4\pi^2 r^2}{GM}$

Open solution

Q80. JEE Main 2024 (Online) 4th April Morning Shift

In an experiment to measure focal length (f) of convex lens, the least counts of the measuring scales for the position of object (u) and for the position of image (v) are Δu and Δv, respectively. The error in the measurement of the focal length of the convex lens will be:

(A) $2f\left[\frac{\Delta \mathrm{u}}{\mathrm{u}} + \frac{\Delta \mathrm{v}}{\mathrm{v}}\right]$

(B) $f\left[\frac{\Delta \mathrm{u}}{\mathrm{u}} + \frac{\Delta \mathrm{v}}{\mathrm{v}}\right]$

(C) $f^2\left[\frac{\Delta \mathrm{u}}{\mathrm{u}^2} + \frac{\Delta \mathrm{v}}{\mathrm{v}^2}\right]$

(D) $\frac{\Delta \mathrm{u}}{\mathrm{u}} + \frac{\Delta \mathrm{v}}{\mathrm{v}}$

Open solution

Q81. JEE Main 2024 (Online) 4th April Morning Shift

The equation of stationary wave is :

$y = 2\mathrm{a}\sin\left(\frac{2\pi \mathrm{nt}}{\lambda}\right)\cos\left(\frac{2\pi x}{\lambda}\right)$.

Which of the following is NOT correct :

(A) The dimensions of nt is [L]

(B) The dimensions of n is $[\mathrm{LT}^{-1}]$

(C) The dimensions of x is [L]

(D) The dimensions of n/λ is [T]

Open solution

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Q82. JEE Main 2024 (Online) 1st February Evening Shift

Open solution

Match **List - I** with **List - II**.

List I (Number)	List II (Significant figure)
(A) 1001	(I) 3
(B) 010.1	(II) 4
(C) 100.100	(III) 5
(D) 0.0010010	(IV) 6

Choose the correct answer from the options given below :

(A) (A)-(II), (B)-(I), (C)-(IV), (D)-(III)

(B) (A)-(IV), (B)-(III), (C)-(I), (D)-(II)

(C) (A)-(I), (B)-(II), (C)-(III), (D)-(IV)

(D) (A)-(III), (B)-(IV), (C)-(II), (D)-(I)

Q83. JEE Main 2024 (Online) 1st February Morning Shift

Open solution

10 divisions on the main scale of a Vernier calliper coincide with 11 divisions on the Vernier scale. If each division on the main scale is of 5 units, the least count of the instrument is :

(A) $\frac{5}{11}$ **(B)** $\frac{10}{11}$ **(C)** $\frac{50}{11}$ **(D)** $\frac{1}{2}$

Q84. JEE Main 2024 (Online) 1st February Morning Shift

Open solution

The dimensional formula of angular impulse is :

(A) $\left[\mathrm{ML}^2\,\mathrm{T}^{-2}\right]$

(B) $\left[\mathrm{ML}^{-2}\,\mathrm{T}^{-1}\right]$

(C) $\left[\mathrm{ML}^2\,\mathrm{T}^{-1}\right]$

(D) $\left[\mathrm{MLT}^{-1}\right]$

Q85. JEE Main 2024 (Online) 1st February Morning Shift

Open solution

The radius (r), length (l) and resistance (R) of a metal wire was measured in the laboratory as

$\mathrm{r} = (0.35 \pm 0.05)\ \mathrm{cm}$

$\mathrm{R} = (100 \pm 10)\ \mathrm{ohm}$

$l = (15 \pm 0.2)\ \mathrm{cm}$

The percentage error in resistivity of the material of the wire is :

(A) 37.3% **(B)** 25.6% **(C)** 35.6% **(D)** 39.9%

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Q86. JEE Main 2024 (Online) 31st January Evening Shift

Consider two physical quantities A and B related to each other as $E = \frac{B-x^2}{At}$ where E, x and t have dimensions of energy, length and time respectively. The dimension of AB is

(A) $L^{-2}M^{1}T^{0}$ (B) $L^{2}M^{-1}T^{1}$ (C) $L^{0}M^{-1}T^{1}$ (D) $L^{-2}M^{-1}T^{1}$

Open solution

Q87. JEE Main 2024 (Online) 31st January Evening Shift

The measured value of the length of a simple pendulum is $20\ \text{cm}$ with $2\ \text{mm}$ accuracy. The time for 50 oscillations was measured to be 40 seconds with 1 second resolution. From these measurements, the accuracy in the measurement of acceleration due to gravity is $\text{N}\%$. The value of N is:

(A) 6 (B) 5 (C) 4 (D) 8

Open solution

Q88. JEE Main 2024 (Online) 31st January Morning Shift

If the percentage errors in measuring the length and the diameter of a wire are 0.1% each. The percentage error in measuring its resistance will be:

(A) 0.144% (B) 0.2% (C) 0.1% (D) 0.3%

Open solution

Q89. JEE Main 2024 (Online) 31st January Morning Shift

A force is represented by $F = ax^2 + bt^{\frac{1}{2}}$

where $x =$ distance and $t =$ time. The dimensions of b^2/a are:

(A) $\left[\text{ML}^2\,\text{T}^{-3}\right]$ (B) $\left[\text{ML}^3\,\text{T}^{-3}\right]$

(C) $\left[MLT^{-2}\right]$ (D) $\left[ML^{-1}T^{-1}\right]$

Open solution

Q90. JEE Main 2024 (Online) 30th January Evening Shift

If 50 Vernier divisions are equal to 49 main scale divisions of a traveling microscope and one smallest reading of main scale is $0.5\ \text{mm}$, the Vernier constant of traveling microscope is

(A) 0.01 mm (B) 0.01 cm (C) 0.1 mm (D) 0.1 cm

Open solution

Q91. JEE Main 2024 (Online) 30th January Evening Shift

If mass is written as $m = k\text{c}^{\text{P}}G^{-1/2}h^{1/2}$ then the value of P will be : (Constants have their usual meaning with k a dimensionless constant)

(A) 1/3 (B) $-1/3$ (C) 1/2 (D) 2

Open solution

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Q92. JEE Main 2024 (Online) 30th January Morning Shift

Open solution

Match List I with List II.

	List I		List II
(A)	Coefficient of viscosity	(I)	$\left[\mathrm{ML}^2\,\mathrm{T}^{-2}\right]$
(B)	Surface tension	(II)	$\left[\mathrm{ML}^2\,\mathrm{T}^{-1}\right]$
(C)	Angular momentum	(III)	$\left[\mathrm{ML}^{-1}\,\mathrm{T}^{-1}\right]$
(D)	Rotational kinetic energy	(IV)	$\left[\mathrm{ML}^0\,\mathrm{T}^{-2}\right]$

Choose the correct answer from the options given below :

(A) (A)-(II), (B)-(I), (C)-(IV), (D)-(III)

(B) (A)-(I), (B)-(II), (C)-(III), (D)-(IV)

(C) (A)-(IV), (B)-(III), (C)-(II), (D)-(I)

(D) (A)-(III), (B)-(IV), (C)-(II), (D)-(I)

Q93. JEE Main 2024 (Online) 29th January Evening Shift

Open solution

A physical quantity Q is found to depend on quantities a, b, c by the relation $Q = \frac{a^4 b^3}{c^2}$. The percentage error in a, b and c are $3\%, 4\%$ and 5% respectively. Then, the percentage error in Q is :

(A) 43%

(B) 34%

(C) 66%

(D) 14%

Q94. JEE Main 2024 (Online) 29th January Morning Shift

Open solution

The resistance $R = \frac{V}{I}$ where $\mathrm{V} = (200 \pm 5)\mathrm{V}$ and $I = (20 \pm 0.2)\mathrm{A}$, the percentage error in the measurement of R is :

(A) 5.5%

(B) 3%

(C) 7%

(D) 3.5%

Q95. JEE Main 2024 (Online) 27th January Evening Shift

Open solution

Given below are two statements : one is labelled as Assertion (A) and the other is labelled as Reason (R).

Assertion (A) : In Vernier calliper if positive zero error exists, then while taking measurements, the reading taken will be more than the actual reading.

Reason (R) : The zero error in Vernier Calliper might have happened due to manufacturing defect or due to rough handling.

In the light of the above statements, choose the correct answer from the options given below :

(A) Both (A) and (R) are correct and (R) is the correct explanation of (A)

(B) Both (A) and (R) are correct but (R) is not the correct explanation of (A)

(C) (A) is true but (R) is false

(D) (A) is false but (R) is true

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Q96. JEE Main 2024 (Online) 27th January Evening Shift

The equation of state of a real gas is given by $\left(\mathrm{P}+\frac{\mathrm{a}}{\mathrm{V}^2}\right)(\mathrm{V}-\mathrm{b})=\mathrm{RT}$, where P, V and T are pressure, volume and temperature respectively and R is the universal gas constant. The dimensions of $\frac{\mathrm{a}}{\mathrm{b}^2}$ is similar to that of :

(A) P (B) RT (C) PV (D) R

Open solution

Q97. JEE Main 2024 (Online) 27th January Morning Shift

Identify the physical quantity that cannot be measured using spherometer :

(A) Radius of curvature of concave surface
(B) Specific rotation of liquids
(C) Thickness of thin plates
(D) Radius of curvature of convex surface

Open solution

Q98. JEE Main 2024 (Online) 27th January Morning Shift

Given below are two statements :

Statement (I) : Planck's constant and angular momentum have same dimensions.

Statement (II) : Linear momentum and moment of force have same dimensions.

In the light of the above statements, choose the correct answer from the options given below :

(A) Statement I is true but Statement II is false
(B) Both Statement I and Statement II are false
(C) Both Statement I and Statement II are true
(D) Statement I is false but Statement II is true

Open solution

Q99. JEE Main 2023 (Online) 15th April Morning Shift

The speed of a wave produced in water is given by $v=\lambda^a g^b \rho^c$. Where λ, g and ρ are wavelength of wave, acceleration due to gravity and density of water respectively. The values of a, b and c respectively, are :

(A) $\frac{1}{2}, 0, \frac{1}{2}$ (B) $1, 1, 0$ (C) $1, -1, 0$ (D) $\frac{1}{2}, \frac{1}{2}, 0$

Open solution

Q100. JEE Main 2023 (Online) 13th April Evening Shift

In the equation $\left[X+\frac{a}{Y^2}\right][Y-b]=\mathrm{R}T$, X is pressure, Y is volume, R is universal gas constant and T is temperature. The physical quantity equivalent to the ratio $\frac{a}{b}$ is:

(A) Impulse
(B) Energy
(C) Pressure gradient
(D) Coefficient of viscosity

Open solution

Q101. JEE Main 2023 (Online) 13th April Morning Shift

A body of mass $(5 \pm 0.5)\ \mathrm{kg}$ is moving with a velocity of $(20 \pm 0.4)\ \mathrm{m/s}$. Its kinetic energy will be

(A) $(1000 \pm 140)\ \mathrm{J}$ (B) $(500 \pm 0.14)\ \mathrm{J}$ (C) $(1000 \pm 0.14)\ \mathrm{J}$ (D) $(500 \pm 140)\ \mathrm{J}$

Open solution

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Q102. JEE Main 2023 (Online) 12th April Morning Shift

Open solution

Match List I with List II

	LIST I		LIST II
A.	Spring constant	I.	$[T^{-1}]$
B.	Angular speed	II.	$[MT^{-2}]$
C.	Angular momentum	III.	$[ML^2]$
D.	Moment of inertia	IV.	$[ML^2T^{-1}]$

Choose the correct answer from the options given below:

(A) A-IV, B-I, C-III, D-II

(B) A-II, B-III, C-I, D-IV

(C) A-II, B-I, C-IV, D-III

(D) A-I, B-III, C-II, D-IV

Q103. JEE Main 2023 (Online) 11th April Evening Shift

Open solution

If force (F), velocity (V) and time (T) are considered as fundamental physical quantities, then dimensional formula of density will be :

(A) $FV^{-2}T^2$

(B) FV^4T^{-6}

(C) $F^2V^{-2}T^6$

(D) $FV^{-4}T^{-2}$

Q104. JEE Main 2023 (Online) 11th April Morning Shift

Open solution

Given below are two statements :

Statements I : Astronomical unit (Au), Parsec (Pc) and Light year (ly) are units for measuring astronomical distances.

Statements II : $\text{Au} < \text{Parsec(Pc)} < \text{ly}$

In the light of the above statements, choose the most appropriate answer from the options given below:

(A) Both Statements I and Statements II are incorrect.

(B) Both Statements I and Statements II are correct.

(C) Statements I is incorrect but Statements II is correct.

(D) Statements I is correct but Statements II is incorrect.

Q105. JEE Main 2023 (Online) 10th April Evening Shift

Open solution

In an experiment with vernier callipers of least count $0.1\ \text{mm}$, when two jaws are joined together the zero of vernier scale lies right to the zero of the main scale and 6th division of vernier scale coincides with the main scale division. While measuring the diameter of a spherical bob, the zero of vernier scale lies in between $3.2\ \text{cm}$ and $3.3\ \text{cm}$ marks, and 4th division of vernier scale coincides with the main scale division. The diameter of bob is measured as

(A) $3.18\ \text{cm}$

(B) $3.22\ \text{cm}$

(C) $3.26\ \text{cm}$

(D) $3.25\ \text{cm}$

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Q106. JEE Main 2023 (Online) 10th April Morning Shift

A physical quantity P is given as

$P = \frac{a^2 b^3}{c\sqrt{d}}$

The percentage error in the measurement of a, b, c and d are 1%, 2%, 3% and 4% respectively. The percentage error in the measurement of quantity P will be

(A) 12% (B) 13% (C) 16% (D) 14%

Open solution

Q107. JEE Main 2023 (Online) 8th April Evening Shift

Match List I with List II

	LIST I		LIST II
A.	Torque	I.	$ML^{-2}T^{-2}$
B.	Stress	II.	$ML^{2}T^{-2}$
C.	Pressure gradient	III.	$ML^{-1}T^{-1}$
D.	Coefficient of viscosity	IV.	$ML^{-1}T^{-2}$

Choose the correct answer from the options given below:

(A) A-II, B-I, C-IV, D-III (B) A-II, B-IV, C-I, D-III

(C) A-IV, B-II, C-III, D-I (D) A-III, B-IV, C-I, D-II

Open solution

Q108. JEE Main 2023 (Online) 8th April Morning Shift

Dimension of $\frac{1}{\mu_0 \epsilon_0}$ should be equal to

(A) T^2/L^2 (B) T/L (C) L^2/T^2 (D) L/T

Open solution

Q109. JEE Main 2023 (Online) 8th April Morning Shift

A cylindrical wire of mass (0.4 ± 0.01)g has length (8 ± 0.04)cm and radius (6 ± 0.03)mm. The maximum error in its density will be:

(A) 1% (B) 5% (C) 4% (D) 3.5%

Open solution

Q110. JEE Main 2023 (Online) 6th April Evening Shift

A 2 meter long scale with least count of 0.2 cm is used to measure the locations of objects on an optical bench. While measuring the focal length of a convex lens, the object pin and the convex lens are placed at 80 cm mark and 1 m mark, respectively. The image of the object pin on the other side of lens coincides with image pin that is kept at 180 cm mark. The $\%$ error in the estimation of focal length is:

(A) 1.70 (B) 0.51 (C) 1.02 (D) 0.85

Open solution

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Q111. JEE Main 2023 (Online) 6th April Morning Shift

Two resistances are given as $R_1 = (10 \pm 0.5)\Omega$ and $R_2 = (15 \pm 0.5)\Omega$. The percentage error in the measurement of equivalent resistance when they are connected in parallel is -

(A) 2.33 (B) 5.33 (C) 4.33 (D) 6.33

Open solution

Q112. JEE Main 2023 (Online) 1st February Evening Shift

If the velocity of light c, universal gravitational constant G and Planck's constant h are chosen as fundamental quantities. The dimensions of mass in the new system is :

(A) $\left[h^1 c^1 G^{-1}\right]$ (B) $\left[h^{-1/2} c^{1/2} G^{1/2}\right]$

(C) $\left[h^{1/2} c^{1/2} G^{-1/2}\right]$ (D) $\left[h^{1/2} c^{-1/2} G^1\right]$

Open solution

Q113. JEE Main 2023 (Online) 1st February Morning Shift

$\left(P + \frac{a}{V^2}\right)(V - b) = RT$ represents the equation of state of some gases. Where P is the pressure, V is the volume, T is the temperature and a, b, R are the constants. The physical quantity, which has dimensional formula as that of $\frac{b^2}{a}$, will be:

(A) Energy density (B) Bulk modulus

(C) Modulus of rigidity (D) Compressibility

Open solution

Q114. JEE Main 2023 (Online) 31st January Evening Shift

Match List **I** with List **II**

LIST I		**LIST II**	
A.	Angular momentum	I.	$\left[ML^2 T^{-2}\right]$
B.	Torque	II.	$\left[ML^{-2} T^{-2}\right]$
C.	Stress	III	$\left[ML^2 T^{-1}\right]$
D.	Pressure gradient	IV.	$\left[ML^{-1} T^{-2}\right]$

Choose the correct answer from the options given below:

(A) A - I, B - IV, C - III, D - II (B) A - III, B - I, C - IV, D - II

(C) A - IV, B - II, C - I, D - III (D) A - II, B - III, C - IV, D - I

Open solution

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Q115. JEE Main 2023 (Online) 30th January Evening Shift

Open solution

Match List **I** with List **II**:

List I		List II	
A.	Torque	I.	$\mathrm{kgm^{-1}\,s^{-2}}$
B.	Energy density	II.	$\mathrm{kg\,ms^{-1}}$
C.	Pressure gradient	III.	$\mathrm{kg\,m^{-2}\,s^{-2}}$
D.	Impulse	IV.	$\mathrm{kg\,m^{2}\,s^{-2}}$

Choose the correct answer from the options given below:

(A) A-IV, B-I, C-III, D-II

(B) A-IV, B-I, C-II, D-III

(C) A-I, B-IV, C-III, D-II

(D) A-IV, B-III, C-I, D-II

Q116. JEE Main 2023 (Online) 29th January Evening Shift

Open solution

The equation of a circle is given by $x^2 + y^2 = a^2$, where a is the radius. If the equation is modified to change the origin other than (0, 0), then find out the correct dimensions of A and B in a new equation : $(x - At)^2 + \left(y - \frac{t}{B}\right)^2 = a^2$. The dimensions of t is given as $[\mathrm{T}^{-1}]$.

(A) $\mathrm{A} = [\mathrm{L}^{-1}\mathrm{T}^{-1}], \mathrm{B} = [\mathrm{LT}^{-1}]$

(B) $\mathrm{A} = [\mathrm{L}^{-1}\mathrm{T}^{-1}], \mathrm{B} = [\mathrm{LT}]$

(C) $\mathrm{A} = [\mathrm{LT}], \mathrm{B} = [\mathrm{L}^{-1}\mathrm{T}^{-1}]$

(D) $\mathrm{A} = [\mathrm{L}^{-1}\mathrm{T}], \mathrm{B} = [\mathrm{LT}^{-1}]$

Q117. JEE Main 2023 (Online) 29th January Morning Shift

Open solution

Match List I with List II :

	List I (Physical Quantity)		List II (Dimensional Formula)
A.	Pressure gradient	I.	$\left[\mathrm{M^0L^2\,T^{-2}}\right]$
B.	Energy density	II.	$\left[\mathrm{M^1L^{-1}\,T^{-2}}\right]$
C.	Electric Field	III.	$\left[\mathrm{M^1L^{-2}\,T^{-2}}\right]$
D.	Latent heat	IV.	$\left[\mathrm{M^1\,L^1\,T^{-3}\,A^{-1}}\right]$

Choose the correct answer from the options given below:

(A) A-III, B-II, C-IV, D-I

(B) A-III, B-II, C-I, D-IV

(C) A-II, B-III, C-IV, D-I

(D) A-II, B-III, C-I, D-IV

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Q118. JEE Main 2023 (Online) 25th January Evening Shift

Open solution

Match List I with List II

	List I		List II
A.	Young's Modulus (Y)	I.	$[ML^{-1}T^{-1}]$
B.	Co-efficient of Viscosity (η)	II.	$[ML^{2}T^{-1}]$
C.	Planck's Constant (h)	III.	$[ML^{-1}T^{-2}]$
D.	Work function (φ)	IV.	$[ML^{2}T^{-2}]$

Choose the correct answer from the options given below :

(A) A-II, B-III, C-IV, D-I

(B) A-I, B-II, C-III, D-IV

(C) A-I, B-III, C-IV, D-II

(D) A-III, B-I, C-II, D-IV

Q119. JEE Main 2023 (Online) 25th January Morning Shift

Open solution

Match List I with List II

	List I		List II
A.	Surface tension	I.	$\mathrm{kg\,m^{-1}\,s^{-1}}$
B.	Pressure	II.	$\mathrm{kg\,ms^{-1}}$
C.	Viscosity	III.	$\mathrm{kg\,m^{-1}\,s^{-2}}$
D.	Impulse	IV.	$\mathrm{kg\,s^{-2}}$

Choose the correct answer from the options given below :

(A) A-IV, B-III, C-I, D-II

(B) A-IV, B-III, C-II, D-I

(C) A-II, B-I, C-III, D-IV

(D) A-III, B-IV, C-I, D-II

Q120. JEE Main 2023 (Online) 24th January Evening Shift

Open solution

The frequency (ν) of an oscillating liquid drop may depend upon radius (r) of the drop, density (ρ) of liquid and the surface tension (s) of the liquid as $\nu = r^a \rho^b s^c$. The values of a, b and c respectively are

(A) $\left(\frac{3}{2}, \frac{1}{2}, -\frac{1}{2}\right)$

(B) $\left(-\frac{3}{2}, -\frac{1}{2}, \frac{1}{2}\right)$

(C) $\left(\frac{3}{2}, -\frac{1}{2}, \frac{1}{2}\right)$

(D) $\left(-\frac{3}{2}, \frac{1}{2}, \frac{1}{2}\right)$

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Q121. JEE Main 2023 (Online) 24th January Morning Shift

Open solution

Match List I with List II

	List-I		List-II
A.	Planck's constant (h)	I.	$[M^1 L^2 T^{-2}]$
B.	Stopping potential (Vs)	II.	$[M^1 L^1 T^{-1}]$
C.	Work function (ϕ)	III.	$[M^1 L^2 T^{-1}]$
D.	Momentum (p)	IV.	$[M^1 L^2 T^{-3} A^{-1}]$

Choose the correct answer from the options given below :

(A) A-I, B-III, C-IV, D-II

(B) A-III, B-IV, C-I, D-II

(C) A-II, B-IV, C-III, D-I

(D) A-III, B-I, C-II, D-IV

Q122. JEE Main 2022 (Online) 30th July Evening Shift

Open solution

If current (I), kinetic energy (K) and charge (Q) are taken as the fundamental quantities, The dimensional representation of power will be :

(A) $[K^{-1} Q I^{-1}]$ **(B)** $[K I Q^{-1}]$ **(C)** $[K Q I^{-1}]$ **(D)** $[K^{-1} Q^{-1} I]$

Q123. JEE Main 2022 (Online) 29th July Evening Shift

Open solution

Match List I with List II.

	List I		List II
A.	Torque	I.	Nms^{-1}
B.	Stress	II.	$J\ kg^{-1}$
C.	Latent Heat	III.	Nm
D.	Power	IV.	Nm^{-2}

Choose the correct answer from the options given below :

(A) A-III, B-II, C-I, D-IV

(B) A-III, B-IV, C-II, D-I

(C) A-IV, B-I, C-III, D-II

(D) A-II, B-III, C-I, D-IV

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Q124. JEE Main 2022 (Online) 29th July Morning Shift

Open solution

Given below are two statements : One is labelled as Assertion (A) and other is labelled as Reason (R).

Assertion (A) : Time period of oscillation of a liquid drop depends on surface tension (S), if density of the liquid is ρ and radius of the drop is r, then $T = K\sqrt{\rho r^3/S^{3/2}}$ is dimensionally correct, where K is dimensionless.

Reason (R) : Using dimensional analysis we get R.H.S. having different dimension than that of time period.

In the light of above statements, choose the correct answer from the options given below.

(A) Both (A) and (R) are true and (R) is the correct explanation of (A)

(B) Both (A) and (R) are true but (R) is not the correct explanation of (A)

(C) (A) is true but (R) is false

(D) (A) is false but (R) is true

Q125. JEE Main 2022 (Online) 29th July Morning Shift

Open solution

A travelling microscope has 20 divisions per cm on the main scale while its vernier scale has total 50 divisions and 25 vernier scale divisions are equal to 24 main scale divisions, what is the least count of the travelling microscope?

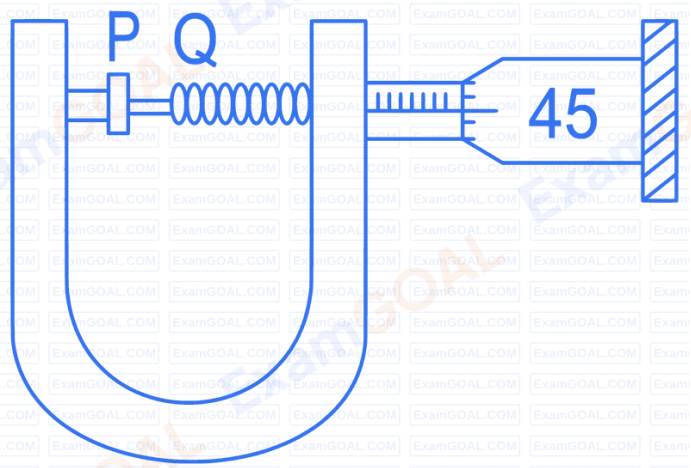
(A) 0.001 cm **(B)** 0.002 mm **(C)** 0.002 cm **(D)** 0.005 cm

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Q126. JEE Main 2022 (Online) 29th July Morning Shift

Open solution

In an experiment to find out the diameter of wire using screw gauge, the following observations were noted :



(A) Screw moves $0.5\ \text{mm}$ on main scale in one complete rotation

(B) Total divisions on circular scale $= 50$

(C) Main scale reading is $2.5\ \text{mm}$

(D) 45th division of circular scale is in the pitch line

(E) Instrument has 0.03 mm negative error

Then the diameter of wire is :

(A) 2.92 mm **(B)** 2.54 mm **(C)** 2.98 mm **(D)** 3.45 mm

Q127. JEE Main 2022 (Online) 28th July Evening Shift

Open solution

Consider the efficiency of Carnot's engine is given by $\eta = \frac{\alpha\beta}{\sin\theta}\log_e \frac{\beta x}{kT}$, where α and β are constants. If T is temperature, k is Boltzmann constant, θ is angular displacement and x has the dimensions of length. Then, choose the incorrect option :

(A) Dimensions of β is same as that of force.

(B) Dimensions of $\alpha^{-1}x$ is same as that of energy.

(C) Dimensions of $\eta^{-1}\sin\theta$ is same as that of $\alpha\beta$.

(D) Dimensions of α is same as that of β.

Q128. JEE Main 2022 (Online) 28th July Morning Shift

Open solution

The dimensions of $\left(\frac{B^2}{\mu_0}\right)$ will be :

(if μ_0 : permeability of free space and B : magnetic field)

(A) $\left[ML^2T^{-2}\right]$ **(B)** $\left[MLT^{-2}\right]$

(C) $\left[ML^{-1}T^{-2}\right]$ **(D)** $\left[ML^2T^{-2}A^{-1}\right]$

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Q129. JEE Main 2022 (Online) 27th July Evening Shift

An expression of energy density is given by $u = \frac{\alpha}{\beta}\sin\left(\frac{\alpha x}{kt}\right)$, where α, β are constants, x is displacement, k is Boltzmann constant and t is the temperature. The dimensions of β will be :

(A) $\left[\mathrm{ML^2\,T^{-2}\theta^{-1}}\right]$ (B) $\left[\mathrm{M^0\,L^2\,T^{-2}}\right]$

(C) $\left[\mathrm{M^0\,L^0\,T^0}\right]$ (D) $\left[\mathrm{M^0\,L^2\,T^0}\right]$

Open solution

Q130. JEE Main 2022 (Online) 27th July Morning Shift

A torque meter is calibrated to reference standards of mass, length and time each with 5% accuracy. After calibration, the measured torque with this torque meter will have net accuracy of :

(A) 15% (B) 25% (C) 75% (D) 5%

Open solution

Q131. JEE Main 2022 (Online) 26th July Evening Shift

In a Vernier Calipers, 10 divisions of Vernier scale are equal to the 9 divisions of main scale. When both jaws of Vernier calipers touch each other, the zero of the Vernier scale is shifted to the left of zero of the main scale and 4th Vernier scale division exactly coincides with the main scale reading. One main scale division is equal to $1\ \mathrm{mm}$. While measuring diameter of a spherical body, the body is held between two jaws. It is now observed that zero of the Vernier scale lies between 30 and 31 divisions of main scale reading and 6th Vernier scale division exactly coincides with the main scale reading. The diameter of the spherical body will be :

(A) 3.02 cm (B) 3.06 cm (C) 3.10 cm (D) 3.20 cm

Open solution

Q132. JEE Main 2022 (Online) 26th July Morning Shift

A screw gauge of pitch $0.5\ \mathrm{mm}$ is used to measure the diameter of uniform wire of length $6.8\ \mathrm{cm}$, the main scale reading is $1.5\ \mathrm{mm}$ and circular scale reading is 7 . The calculated curved surface area of wire to appropriate significant figures is :

[Screw gauge has 50 divisions on its circular scale]

(A) 6.8 cm^2 (B) 3.4 cm^2 (C) 3.9 cm^2 (D) 2.4 cm^2

Open solution

Q133. JEE Main 2022 (Online) 25th July Evening Shift

The maximum error in the measurement of resistance, current and time for which current flows in an electrical circuit are $1\%, 2\%$ and 3% respectively. The maximum percentage error in the detection of the dissipated heat will be :

(A) 2 (B) 4 (C) 6 (D) 8

Open solution

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Q134. JEE Main 2022 (Online) 25th July Morning Shift

If momentum [P], area $[\mathrm{A}]$ and time $[\mathrm{T}]$ are taken as fundamental quantities, then the dimensional formula for coefficient of viscosity is :

(A) $\left[\mathrm{P\,A^{-1}\,T^{0}}\right]$ (B) $\left[\mathrm{P\,A\,T^{-1}}\right]$

(C) $\left[\mathrm{P\,A^{-1}\,T}\right]$ (D) $\left[\mathrm{P\,A^{-1}\,T^{-1}}\right]$

Open solution

Q135. JEE Main 2022 (Online) 25th July Morning Shift

Which of the following physical quantities have the same dimensions?

(A) Electric displacement $(\vec{\mathrm{D}})$ and surface charge density

(B) Displacement current and electric field

(C) Current density and surface charge density

(D) Electric potential and energy

Open solution

Q136. JEE Main 2022 (Online) 30th June Morning Shift

If n main scale divisions coincide with (n + 1) vernier scale divisions. The least count of vernier callipers, when each centimetre on the main scale is divided into five equal parts, will be :

(A) $\frac{2}{n+1}$ mm (B) $\frac{5}{n+1}$ mm (C) $\frac{1}{2n}$ mm (D) $\frac{1}{5n}$ mm

Open solution

Q137. JEE Main 2022 (Online) 29th June Morning Shift

In van der Waal equation $\left[P+\frac{a}{V^2}\right]$ [V − b] = RT; P is pressure, V is volume, R is universal gas constant and T is temperature. The ratio of constants $\frac{a}{b}$ is dimensionally equal to :

(A) $\frac{P}{V}$ (B) $\frac{V}{P}$ (C) PV (D) PV^3

Open solution

Q138. JEE Main 2022 (Online) 28th June Evening Shift

Velocity (v) and acceleration (a) in two systems of units 1 and 2 are related as $v_2=\frac{n}{m^2}v_1$ and $a_2=\frac{a_1}{mn}$ respectively. Here m and n are constants. The relations for distance and time in two systems respectively are :

(A) $\frac{n^3}{m^3}L_1=L_2$ and $\frac{n^2}{m}T_1=T_2$ (B) $L_1=\frac{n^4}{m^2}L_2$ and $T_1=\frac{n^2}{m}T_2$

(C) $L_1=\frac{n^2}{m}L_2$ and $T_1=\frac{n^4}{m^2}T_2$ (D) $\frac{n^2}{m}L_1=L_2$ and $\frac{n^4}{m^2}T_1=T_2$

Open solution

Q139. JEE Main 2022 (Online) 27th June Evening Shift

The SI unit of a physical quantity is pascal-second. The dimensional formula of this quantity will be :

(A) $[ML^{-1}T^{-1}]$ (B) $[ML^{-1}T^{-2}]$ (C) $[ML^{2}T^{-1}]$ (D) $[M^{-1}L^{3}T^{0}]$

Open solution

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Q140. JEE Main 2022 (Online) 27th June Morning Shift

A silver wire has a mass (0.6 ± 0.006) g, radius (0.5 ± 0.005) mm and length (4 ± 0.04) cm. The maximum percentage error in the measurement of its density will be :

(A) 4% (B) 3% (C) 6% (D) 7%

Open solution

Q141. JEE Main 2022 (Online) 26th June Evening Shift

The dimension of mutual inductance is :

(A) $[ML^2T^{-2}A^{-1}]$ (B) $[ML^2T^{-3}A^{-1}]$ (C) $[ML^2T^{-2}A^{-2}]$ (D) $[ML^2T^{-3}A^{-2}]$

Open solution

Q142. JEE Main 2022 (Online) 26th June Morning Shift

An expression for a dimensionless quantity P is given by $P = \frac{\alpha}{\beta}\log_e\left(\frac{kt}{\beta x}\right)$; where α and β are constants, x is distance; k is Boltzmann constant and t is the temperature. Then the dimensions of α will be :

(A) $[M^0\,L^{-1}\,T^0]$ (B) $[M\,L^0\,T^{-2}]$ (C) $[M\,L\,T^{-2}]$ (D) $[M\,L^2\,T^{-2}]$

Open solution

Q143. JEE Main 2022 (Online) 25th June Morning Shift

If $Z = \frac{A^2B^3}{C^4}$, then the relative error in Z will be :

(A) $\frac{\Delta A}{A} + \frac{\Delta B}{B} + \frac{\Delta C}{C}$ (B) $\frac{2\Delta A}{A} + \frac{3\Delta B}{B} - \frac{4\Delta C}{C}$

(C) $\frac{2\Delta A}{A} + \frac{3\Delta B}{B} + \frac{4\Delta C}{C}$ (D) $\frac{\Delta A}{A} + \frac{\Delta B}{B} - \frac{\Delta C}{C}$

Open solution

Q144. JEE Main 2022 (Online) 24th June Evening Shift

Identify the pair of physical quantities that have same dimensions:

(A) velocity gradient and decay constant (B) Wien's constant and Stefan constant

(C) angular frequency and angular momentum (D) wave number and Avogadro number

Open solution

Q145. JEE Main 2022 (Online) 24th June Morning Shift

Identify the pair of physical quantities which have different dimensions:

(A) Wave number and Rydberg's constant (B) Stress and Coefficient of elasticity

(C) Coercivity and Magnetisation (D) Specific heat capacity and Latent heat

Open solution

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Q146. JEE Main 2021 (Online) 1st September Evening Shift

A student determined Young's Modulus of elasticity using the formula $Y = \frac{MgL^3}{4bd^3\delta}$. The value of g is taken to be 9.8 m/s^2, without any significant error, his observations are as follows.

Open solution

Physical Quantity	Least count of the Equipment used for measurement	Observed value
Mass (M)	1 g	2 kg
Length of bar (L)	1 mm	1 m
Breadth of bar (b)	0.1 mm	4 cm
Thickness of bar (d)	0.01 mm	0.4 cm
Depression (δ)	0.01 mm	5 mm

Then the fractional error in the measurement of Y is :

(A) 0.0083 (B) 0.0155 (C) 0.155 (D) 0.083

Q147. JEE Main 2021 (Online) 1st September Evening Shift

Two resistors $R_1 = (4 \pm 0.8)\ \Omega$ and $R_2 = (4 \pm 0.4)\ \Omega$ are connected in parallel. The equivalent resistance of their parallel combination will be :

(A) $(4 \pm 0.4)\ \Omega$ (B) $(2 \pm 0.4)\ \Omega$ (C) $(2 \pm 0.3)\ \Omega$ (D) $(4 \pm 0.3)\ \Omega$

Open solution

Q148. JEE Main 2021 (Online) 31st August Evening Shift

If velocity [V], time [T] and force [F] are chosen as the base quantities, the dimensions of the mass will be :

(A) $[FT^{-1}\ V^{-1}]$ (B) $[FTV^{-1}]$ (C) $[FT^2\ V]$ (D) $[FVT^{-1}]$

Open solution

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Q149. JEE Main 2021 (Online) 31st August Morning Shift

Open solution

Match List - I with List - II.

	List - I		List - II
(a)	Torque	(i)	MLT^{-1}
(b)	Impulse	(ii)	MT^{-2}
(c)	Tension	(iii)	ML^2T^{-2}
(d)	Surface Tension	(iv)	MLT^{-2}

Choose the most appropriate answer from the options given below :

(A) (a)-(iii), (b)-(i), (c)-(iv), (d)-(ii)

(B) (a)-(ii), (b)-(i), (c)-(iv), (d)-(iii)

(C) (a)-(i), (b)-(iii), (c)-(iv), (d)-(ii)

(D) (a)-(iii), (b)-(iv), (c)-(i), (d)-(ii)

Q150. JEE Main 2021 (Online) 31st August Morning Shift

Open solution

Which of the following equations is dimensionally incorrect?

Where t = time, h = height, s = surface tension, θ = angle, ρ = density, a, r = radius, g = acceleration due to gravity, v = volume, p = pressure, W = work done, Γ = torque, $\in$ = permittivity, E = electric field, J = current density, L = length.

(A) $v = \frac{\pi p a^4}{8\eta L}$

(B) $h = \frac{2s\cos\theta}{\rho r g}$

(C) $J = \in \frac{\partial E}{\partial t}$

(D) $W = \Gamma\theta$

Q151. JEE Main 2021 (Online) 27th August Evening Shift

Open solution

Match List - I with List - II.

	List - I		List - II
(a)	R_H (Rydberg constant)	(i)	$kg\,m^{-1}s^{-1}$
(b)	h (Planck's constant)	(ii)	$kg\,m^2s^{-1}$
(c)	μ_B (Magnetic field energy density)	(iii)	m^{-1}
(d)	η (coefficient of viscocity)	(iv)	$kg\,m^{-1}s^{-2}$

Choose the most appropriate answer from the options given below :

(A) (a)-(ii), (b)-(iii), (c)-(iv), (d)-(i)

(B) (a)-(iii), (b)-(ii), (c)-(iv), (d)-(i)

(C) (a)-(iv), (b)-(ii), (c)-(i), (d)-(iii)

(D) (a)-(iii), (b)-(ii), (c)-(i), (d)-(iv)

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Q152. JEE Main 2021 (Online) 27th August Evening Shift

If force (F), length (L) and time (T) are taken as the fundamental quantities. Then what will be the dimension of density :

(A) $[FL^{-4}T^2]$ (B) $[FL^{-3}T^2]$ (C) $[FL^{-5}T^2]$ (D) $[FL^{-3}T^3]$

Open solution

Q153. JEE Main 2021 (Online) 27th August Morning Shift

Which of the following is not a dimensionless quantity?

(A) Relative magnetic permeability (μ_r)
(B) Power factor
(C) Permeability of free space (μ_0)
(D) Quality factor

Open solution

Q154. JEE Main 2021 (Online) 27th August Morning Shift

If E and H represents the intensity of electric field and magnetising field respectively, then the unit of E/H will be :

(A) ohm (B) mho (C) joule (D) newton

Open solution

Q155. JEE Main 2021 (Online) 26th August Evening Shift

Match List - I with List - II

List-I		List-II	
(a)	Magnetic Induction	(i)	$ML^2T^{-2}A^{-1}$
(b)	Magnetic Flux	(ii)	$M^0L^{-1}A$
(c)	Magnetic Permeability	(iii)	$MT^{-2}A^{-1}$
(d)	Magnetization	(iv)	$MLT^{-2}A^{-2}$

Open solution

Choose the most appropriate answer from the options given below :

(A) (a)-(ii), (b)-(iv), (c)-(i), (d)-(iii)
(B) (a)-(ii), (b)-(i), (c)-(iv), (d)-(iii)
(C) (a)-(iii), (b)-(ii), (c)-(iv), (d)-(i)
(D) (a)-(iii), (b)-(i), (c)-(iv), (d)-(ii)

Q156. JEE Main 2021 (Online) 26th August Evening Shift

If the length of the pendulum in pendulum clock increases by 0.1%, then the error in time per day is :

(A) 86.4 s (B) 4.32 s (C) 43.2 s (D) 8.64 s

Open solution

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Q157. JEE Main 2021 (Online) 26th August Morning Shift

In a Screw Gauge, fifth division of the circular scale coincides with the reference line when the ratchet is closed. There are 50 divisions on the circular scale, and the main scale moves by 0.5 mm on a complete rotation. For a particular observation the reading on the main scale is 5 mm and the 20th division of the circular scale coincides with reference line. Calculate the true reading.

Open solution

(A) 5.00 mm (B) 5.25 mm (C) 5.15 mm (D) 5.20 mm

Q158. JEE Main 2021 (Online) 26th August Morning Shift

If E, L, M and G denote the quantities as energy, angular momentum, mass and constant of gravitation respectively, then the dimensions of P in the formula $P = EL^2M^{-5}G^{-2}$ are :

Open solution

(A) $[M^0 L^1 T^0]$ (B) $[M^{-1} L^{-1} T^2]$ (C) $[M^1 L^1 T^{-2}]$ (D) $[M^0 L^0 T^0]$

Q159. JEE Main 2021 (Online) 27th July Evening Shift

Match List I with List II.

Open solution

	List - I		List - II
(a)	Capacitance, C	(i)	$M^1L^1T^{-3}A^{-1}$
(b)	Permittivity of free space, ε_0	(ii)	$M^{-1}L^{-3}T^4A^2$
(c)	Permeability of free space, μ_0	(iii)	$M^{-1}L^{-2}T^4A^2$
(d)	Electric field, E	(iv)	$M^1L^1T^{-2}A^{-2}$

Choose the correct answer from the options given below

(A) (a) $\rightarrow$ (iii), (b) $\rightarrow$ (ii), (c) $\rightarrow$ (iv), (d) $\rightarrow$ (i)

(B) (a) $\rightarrow$ (iii), (b) $\rightarrow$ (iv), (c) $\rightarrow$ (ii), (d) $\rightarrow$ (i)

(C) (a) $\rightarrow$ (iv), (b) $\rightarrow$ (ii), (c) $\rightarrow$ (iii), (d) $\rightarrow$ (i)

(D) (a) $\rightarrow$ (iv), (b) $\rightarrow$ (iii), (c) $\rightarrow$ (ii), (d) $\rightarrow$ (i)

Q160. JEE Main 2021 (Online) 27th July Evening Shift

A physical quantity 'y' is represented by the formula $y = m^2r^{-4}g^x l^{-\frac{3}{2}}$

If the percentage errors found in y, m, r, l and g are 18, 1, 0.5, 4 and p respectively, then find the value of x and p.

Open solution

(A) 5 and ± 2 (B) 4 and ± 3

(C) $\frac{16}{3}$ and $\pm\frac{3}{2}$ (D) 8 and $\pm$ 2

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Q161. JEE Main 2021 (Online) 27th July Morning Shift

Open solution

Assertion A : If in five complete rotations of the circular scale, the distance travelled on main scale of the screw gauge is 5 mm and there are 50 total divisions on circular scale, then least count is 0.001 cm.

Reason R :

Least Count = $\frac{Pitch}{Total\ divisions\ on\ circular\ scale}$

In the light of the above statements, choose the most appropriate answer from the options given below :

(A) A is not correct but R is correct.

(B) Both A and R are correct and R is the correct explanation of A.

(C) A is correct but R is not correct.

(D) Both A and R are correct and R is NOT the correct explanation of A.

Q162. JEE Main 2021 (Online) 25th July Evening Shift

Open solution

The force is given in terms of time t and displacement x by the equation

F = A cos Bx + C sin Dt

The dimensional formula of $\frac{AD}{B}$ is :

(A) $[M^0LT^{-1}]$ (B) $[ML^2T^{-3}]$ (C) $[M^1L^1T^{-2}]$ (D) $[M^2L^2T^{-3}]$

Q163. JEE Main 2021 (Online) 20th July Evening Shift

Open solution

If time (t), velocity (v), and angular momentum (l) are taken as the fundamental units. Then the dimension of mass (m) in terms of t, v and l is :

(A) $[t^{-1}v^1l^{-2}]$ (B) $[t^1v^2l^{-1}]$ (C) $[t^{-2}v^{-1}l^1]$ (D) $[t^{-1}v^{-2}l^1]$

Q164. JEE Main 2021 (Online) 20th July Morning Shift

Open solution

The entropy of any system is given by
$S = \alpha^2\beta \ln\left[\frac{\mu kR}{J\beta^2} + 3\right]$ where α and β are the constants. μ, J, k and R are no. of moles, mechanical equivalent of heat, Boltzmann constant and gas constant respectively.
[Take $S = \frac{dQ}{T}$]
Choose the incorrect option from the following :

(A) α and J have the same dimensions.

(B) S and α have different dimensions

(C) S, β, k and μR have the same dimensions

(D) α and k have the same dimensions

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Q165. JEE Main 2021 (Online) 18th March Morning Shift

In the experiment of Ohm's law, a potential difference of 5.0 V is applied across the end of a conductor of length 10.0 cm and diameter of 5.00 mm. The measured current in the conductor is 2.00 A. The maximum permissible percentage error in the resistivity of the conductor is :

Open solution

(A) 3.9 (B) 8.4 (C) 7.5 (D) 3.0

Q166. JEE Main 2021 (Online) 18th March Morning Shift

The time period of a simple pendulum is given by $T = 2\pi\sqrt{\frac{l}{g}}$. The measured value of the length of pendulum is 10 cm known to a 1mm accuracy. The time for 200 oscillations of the pendulum is found to be 100 second using a clock of 1s resolution. The percentage accuracy in the determination of 'g' using this pendulum is 'x'. The value of 'x' to be nearest integer is :-

Open solution

(A) 2% (B) 3% (C) 5% (D) 4%

Q167. JEE Main 2021 (Online) 17th March Morning Shift

The vernier scale used for measurement has a positive zero error of 0.2 mm. If while taking a measurement it was noted that '0' on the vernier scale lies between 8.5 cm and 8.6 cm, vernier coincidence is 6, then the correct value of measurement is _____ cm. (least count = 0.01 cm)

Open solution

(A) 8.58 cm (B) 8.54 cm (C) 8.56 cm (D) 8.36 cm

Q168. JEE Main 2021 (Online) 16th March Evening Shift

In order to determine the Young's Modulus of a wire of radius 0.2 cm (measured using a scale of least count = 0.001 cm) and length 1m (measured using a scale of least count = 1 mm), a weight of mass 1 kg (measured using a scale of least count = 1 g) was hanged to get the elongation of 0.5 cm (measured using a scale of least count 0.001 cm). What will be the fractional error in the value of Young's Modulus determined by this experiment?

Open solution

(A) 0.14% (B) 9% (C) 1.4% (D) 0.9%

Q169. JEE Main 2021 (Online) 16th March Morning Shift

One main scale division of a vernier callipers is 'a' cm and n^{th} division of the vernier scale coincide with $(n-1)^{th}$ division of the main scale. The least count of the callipers in mm is :

Open solution

(A) $\frac{10a}{n}$ (B) $\frac{10na}{(n-1)}$

(C) $\left(\frac{n-1}{10n}\right)a$ (D) $\frac{10a}{(n-1)}$

Q170. JEE Main 2021 (Online) 26th February Evening Shift

If 'C' and 'V' represent capacitance and voltage respectively then what are the dimensions of λ where C/V = λ ?

Open solution

(A) $[M^{-3}L^{-4}I^{3}T^{7}]$ (B) $[M^{-2}L^{-3}I^{2}T^{6}]$

(C) $[M^{-2}L^{-4}I^{3}T^{7}]$ (D) $[M^{-1}L^{-3}I^{-2}T^{-7}]$

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Q171. JEE Main 2021 (Online) 26th February Morning Shift

Open solution

In a typical combustion engine the workdone by a gas molecule is given by $W = \alpha^2 \beta e^{\frac{-\beta x^2}{kT}}$, where x is the displacement, k is the Boltzmann constant and T is the temperature. If α and β are constants, dimensions of α will be :

(A) $[M^0LT^0]$ (B) $[MLT^{-1}]$ (C) $[MLT^{-2}]$ (D) $[M^2LT^{-2}]$

Q172. JEE Main 2021 (Online) 25th February Evening Shift

Open solution

If e is the electronic charge, c is the speed of light in free space and h is Planck's constant, the quantity $\frac{1}{4\pi\varepsilon_0}\frac{|e|^2}{hc}$ has dimensions of :

(A) $[MLT^{-1}]$ (B) $[MLT^0]$ (C) $[M^0L^0T^0]$ (D) $[LC^{-1}]$

Q173. JEE Main 2021 (Online) 25th February Morning Shift

Open solution

Match List - I with List - II :

	List I		List II
(a)	h (Planck's constant)	(i)	$[MLT^{-1}]$
(b)	E (kinetic energy)	(ii)	$[ML^2T^{-1}]$
(c)	V (electric potential)	(iii)	$[ML^2T^{-2}]$
(d)	P (linear momentum)	(iv)	$[ML^2I^{-1}T^{-3}]$

Choose the correct answer from the options given below :

(A) (a) $\rightarrow$ (ii), (b) $\rightarrow$ (iii), (c) $\rightarrow$ (iv), (d) $\rightarrow$ (i)

(B) (a) $\rightarrow$ (i), (b) $\rightarrow$ (ii), (c) $\rightarrow$ (iv), (d) $\rightarrow$ (iii)

(C) (a) $\rightarrow$ (iii), (b) $\rightarrow$ (ii), (c) $\rightarrow$ (iv), (d) $\rightarrow$ (i)

(D) (a) $\rightarrow$ (iii), (b) $\rightarrow$ (iv), (c) $\rightarrow$ (ii), (d) $\rightarrow$ (i)

Q174. JEE Main 2021 (Online) 25th February Morning Shift

Open solution

The pitch of the screw gauge is 1 mm and there are 100 divisions on the circular scale. When nothing is put in between the jaws, the zero of the circular scale lies 8 divisions below the reference line. When a wire is placed between the jaws, the first linear scale division is clearly visible while 72nd division on circular scale coincides with the reference line. The radius of the wire is :

(A) 1.80 mm (B) 0.90 mm (C) 0.82 mm (D) 1.64 mm

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Q175. JEE Main 2021 (Online) 24th February Evening Shift

Open solution

The period of oscillation of a simple pendulum is $T = 2\pi\sqrt{\frac{L}{g}}$. Measured value of 'L' is 1.0 m from meter scale having a minimum division of 1 mm and time of one complete oscillation is 1.95 s measured from stopwatch of 0.01 s resolution. The percentage error in the determination of 'g' will be :

(A) 1.30% (B) 1.33% (C) 1.13% (D) 1.03%

Q176. JEE Main 2021 (Online) 24th February Morning Shift

Open solution

The work done by a gas molecule in an isolated system is

given by, $W = \alpha\beta^2 e^{-\frac{x^2}{\alpha kT}}$, where x is the displacement, k is the Boltzmann constant and T is the temperature. α and β are constants. Then the dimensions of β will be :

(A) $[M^0LT^0]$ (B) $[ML^2T^{-2}]$ (C) $[MLT^{-2}]$ (D) $[M^2LT^2]$

Q177. JEE Main 2020 (Online) 6th September Evening Slot

Open solution

A student measuring the diameter of a pencil of circular cross-section with the help of a vernier scale records the following four readings 5.50 mm, 5.55 mm, 5.45 mm, 5.65 mm. The average of these four readings is 5.5375 mm and the standard deviation of the data is 0.07395 mm. The average diameter
of the pencil should therefore be recorded as :

(A) (5.54 ± 0.07) mm (B) (5.5375 ± 0.0740) mm
(C) (5.5375 ± 0.0739) mm (D) (5.538 ± 0.074) mm

Q178. JEE Main 2020 (Online) 6th September Morning Slot

Open solution

A screw gauge has 50 divisions on its circular scale. The circular scale is 4 units ahead of the pitch scale marking, prior to use. Upon one complete rotation of the circular scale, a displacement of 0.5mm is noticed on the pitch scale. The nature of zero error involved and the least count of the screw gauge, are respectively :

(A) Positive, 0.1 mm (B) Positive, 0.1 μm
(C) Positive, 10 μm (D) Negative, 2 μm

Q179. JEE Main 2020 (Online) 5th September Evening Slot

Open solution

The quantities x = $\frac{1}{\sqrt{\mu_0\varepsilon_0}}$, y = $\frac{E}{B}$ and z = $\frac{l}{CR}$ are
defined where C-capacitance, R-Resistance, l-length, E-Electric field, B-magnetic field and ε_0, μ_0, - free space permittivity and permeability respectively. Then :

(A) Only y and z have the same dimension (B) x, y and z have the same dimension
(C) Only x and y have the same dimension (D) Only x and z have the same dimension

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Q180. JEE Main 2020 (Online) 5th September Morning Slot

A physical quantity z depends on four observables a, b, c and d, as z = $\frac{a^2 b^{\frac{2}{3}}}{\sqrt{c}d^3}$. The percentages of error in the measurement of a, b, c and d are 2%, 1.5%, 4% and 2.5% respectively. The percentage of error in z is :

Open solution

(A) 13.5 %
(B) 14.5%
(C) 16.5%
(D) 12.25%

Q181. JEE Main 2020 (Online) 4th September Evening Slot

A quantity x is given by $\left(\frac{IFv^2}{WL^4}\right)$ in terms of moment of inertia I, force F, velocity v, work W and Length L. The dimensional formula for x is same as that of :

Open solution

(A) Coefficient of viscosity
(B) Force constant
(C) Energy density
(D) Planck's constant

Q182. JEE Main 2020 (Online) 4th September Morning Slot

Dimensional formula for thermal conductivity is (here K denotes the temperature):

Open solution

(A) $MLT^{-3}K^{-1}$
(B) $MLT^{-2}K^{-2}$
(C) $MLT^{-2}K$
(D) $MLT^{-3}K$

Q183. JEE Main 2020 (Online) 3rd September Evening Slot

Amount of solar energy received on the earth's surface per unit area per unit time is defined as solar constant. Dimension of solar constant is :

Open solution

(A) MLT^{-2}
(B) ML^0T^{-3}
(C) $M^2L^0T^{-1}$
(D) ML^2T^{-2}

Q184. JEE Main 2020 (Online) 3rd September Morning Slot

Using screw gauge of pitch 0.1 cm and 50 divisions on its circular scale, the thickness of an object is measured. It should correctly be recorded as

Open solution

(A) 2.123 cm
(B) 2.124 cm
(C) 2.125 cm
(D) 2.121 cm

Q185. JEE Main 2020 (Online) 2nd September Evening Slot

If momentum (P), area (A) and time (T) are taken to be the fundamental quantities then the dimensional formula for energy is

Open solution

(A) $[P^2AT^{-2}]$
(B) $\left[P^{\frac{1}{2}}AT^{-1}\right]$
(C) $\left[PA^{\frac{1}{2}}T^{-1}\right]$
(D) $[PA^{-1}T^{-2}]$

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Q186. JEE Main 2020 (Online) 2nd September Morning Slot

Open solution

The least count of the main scale of a vernier callipers is 1 mm. Its vernier scale is divided into 10 divisions and coincide with 9 divisions of the main scale. When jaws are touching each other, the 7th division of vernier scale coincides with a division of main scale and the zero of vernier scale is lying right side of the zero of main scale. When this vernier is used to measure length of a cylinder the zero of the vernier scale between 3.1 cm and 3.2 cm and 4th VSD coincides with a main scale division. The length of the cylinder is : (VSD is vernier scale division)

(A) 3.21 cm (B) 2.99 cm (C) 3.07 cm (D) 3.2 cm

Q187. JEE Main 2020 (Online) 2nd September Morning Slot

Open solution

If speed V, area A and force F are chosen as fundamental units, then the dimension of Young's modulus will be

(A) $FA^{-1}V^{0}$ (B) $FA^{2}V^{-1}$ (C) $FA^{2}V^{-2}$ (D) $FA^{2}V^{-3}$

Q188. JEE Main 2020 (Online) 9th January Evening Slot

Open solution

For the four sets of three measured physical quantities as given below. Which of the following options is correct ?

(i) $A_1 = 24.36, B_1 = 0.0724, C_1 = 256.2$

(ii) $A_2 = 24.44, B_2 = 16.082, C_2 = 240.2$

(iii) $A_3 = 25.2, B_3 = 19.2812, C_3 = 236.183$

(iv) $A_4 = 25, B_4 = 236.191, C_4 = 19.5$

(A) $A_1 + B_1 + C_1 = A_2 + B_2 + C_2 = A_3 + B_3 + C_3 = A_4 + B_4 + C_4$

(B) $A_4 + B_4 + C_4 < A_1 + B_1 + C_1 < A_3 + B_3 + C_3 < A_2 + B_2 + C_2$

(C) $A_4 + B_4 + C_4 < A_1 + B_1 + C_1 = A_2 + B_2 + C_2 = A_3 + B_3 + C_3$

(D) $A_4 + B_4 + C_4 > A_3 + B_3 + C_3 = A_2 + B_2 + C_2 > A_1 + B_1 + C_1$

Q189. JEE Main 2020 (Online) 9th January Morning Slot

Open solution

A quantity f is given by $f = \sqrt{\frac{hc^5}{G}}$ where c is speed of light, G universal gravitational constant and h is the Planck's constant. Dimension of f is that of :

(A) Energy (B) Momentum (C) Area (D) Volume

Q190. JEE Main 2020 (Online) 9th January Morning Slot

Open solution

If the screw on a screw-gauge is given six rotations, it moves by 3 mm on the main scale. If there are 50 divisions on the circular scale the least count of the screw gauge is :

(A) 0.001 mm (B) 0.01 cm (C) 0.02 mm (D) 0.001 cm

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Q191. JEE Main 2020 (Online) 8th January Evening Slot

A simple pendulum is being used to determine the value of gravitational acceleration g at a certain place. The length of the pendulum is 25.0 cm and a stop watch with 1s resolution measures the time taken for 40 oscillations to be 50 s. The accuracy in g is :

Open solution

(A) 4.40% (B) 3.40% (C) 2.40% (D) 5.40%

Q192. JEE Main 2020 (Online) 8th January Morning Slot

The dimension of stopping potential V_0 in photoelectric effect in units of Planck's constant 'h', speed of light 'c' and Gravitational constant 'G' and ampere A is :

Open solution

(A) $h^{1/3} G^{2/3} c^{1/3} A^{-1}$ (B) $h^0 c^5 G^{-1} A^{-1}$

(C) $h^{2/3} c^{5/3} G^{1/3} A^{-1}$ (D) $h^2 G^{3/2} c^{1/3} A^{-1}$

Q193. JEE Main 2020 (Online) 7th January Evening Slot

The dimension of $\frac{B^2}{2\mu_0}$, where B is magnetic field and μ_0 is the magnetic permeability of vacuum, is :

Open solution

(A) ML^2T^{-2} (B) MLT^{-2} (C) $ML^{-1}T^{-2}$ (D) ML^2T^{-1}

Q194. JEE Main 2019 (Online) 12th April Morning Slot

Which of the following combinations has the dimension of electrical resistance ($\in_0$ is the permittivity of vacuum and μ_0 is the permeability of vacuum)?

Open solution

(A) $\sqrt{\frac{\in_0}{\mu_0}}$ (B) $\frac{\in_0}{\mu_0}$

(C) $\sqrt{\frac{\mu_0}{\in_0}}$ (D) $\frac{\mu_0}{\in_0}$

Q195. JEE Main 2019 (Online) 10th April Evening Slot

In the formula $X = 5YZ^2$, X and Z have dimensions of capacitance and magnetic field, respectively. What are the dimensions of Y in SI units?

Open solution

(A) $[M^{-3}L^{-2}T^8A^4]$ (B) $[M^{-2}L^{-2}T^6A^3]$ (C) $[M^{-1}L^{-2}T^4A^2]$ (D) $[M^{-2}L^0T^{-4}A^{-2}]$

Q196. JEE Main 2019 (Online) 9th April Evening Slot

The area of a square is 5.29 cm^2. The area of 7 such squares taking into account the significant figures is :-

Open solution

(A) 37.0 cm^2 (B) 37 cm^2 (C) 37.030 cm^2 (D) 37.03 cm^2

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Q197. JEE Main 2019 (Online) 9th April Morning Slot

In the density measurement of a cube, the mass and edge length are measured as (10.00 ± 0.10) kg and (0.10 ± 0.01) m, respectively. The error in the measurement of density is :

(A) 0.01 kg/m^3 (B) 0.10 kg/m^3 (C) 0.31 kg/m^3 (D) 0.07 kg/m^3

Open solution

Q198. JEE Main 2019 (Online) 8th April Evening Slot

In a simple pendulum experiment for determination of acceleration due to gravity (g), time taken for 20 oscillations is measured by using a watch of 1 second least count. The mean value of time taken comes out to be 30 s. The length of pendulum is measured by using a meter scale of least count 1 mm and the value obtained is 55.0 cm. The percentage error in the determination of g is close to :-

(A) 0.2% (B) 3.5% (C) 0.7% (D) 6.8%

Open solution

Q199. JEE Main 2019 (Online) 8th April Evening Slot

If surface tension (S), Moment of inertia (I) and Planck's constant (h), were to be taken as the fundamental units, the dimensional formula for linear momentum would be :-

(A) $S^{1/2}I^{1/2}h^0$ (B) $S^{3/2}I^{1/2}h^0$ (C) $S^{1/2}I^{1/2}h^{-1}$ (D) $S^{1/2}I^{3/2}h^{-1}$

Open solution

Q200. JEE Main 2019 (Online) 8th April Morning Slot

In SI units, the dimensions of $\sqrt{\frac{\in_0}{\mu_0}}$ is :

(A) $A^{-1}TML^3$ (B) $A^2T^3M^{-1}L^{-2}$ (C) $AT^{-3}ML^{3/2}$ (D) $AT^2M^{-1}L^{-1}$

Open solution

Q201. JEE Main 2019 (Online) 12th January Evening Slot

Let ℓ, r, C and V represent inductance, resistance, capacitance and voltage, respectively. The dimension of $\frac{\ell}{rCV}$ in SI units will be :

(A) $[A^{-1}]$ (B) [LTA] (C) $[LA^{-2}]$ (D) $[LT^2]$

Open solution

Q202. JEE Main 2019 (Online) 12th January Morning Slot

The least count of the main scale of a screw gauge is 1 mm. The minimum number of divisions on its circular scale required to measure 5 μm diameter of a wire is :

(A) 500 (B) 100 (C) 200 (D) 50

Open solution

Q203. JEE Main 2019 (Online) 11th January Evening Slot

If speed (V), acceleration (A) and force (F) are considered as fundamental units, the dimension of Young's modulus will be:

(A) $V^{-2}A^2F^2$ (B) $V^{-4}A^{-2}F$ (C) $V^{-4}A^2F$ (D) $V^{-2}A^2F^{-2}$

Open solution

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Q204. JEE Main 2019 (Online) 11th January Morning Slot

The force of interaction between two atoms is given by F = $\alpha\beta\exp\left(-\frac{x^2}{\alpha kT}\right)$; where x is the distance, k is the Boltzmann constant and T is temperature and α and β are two constants. The dimension of β is :

Open solution

(A) $M^2L^2T^{-2}$ (B) M^2LT^{-4} (C) MLT^{-4} (D) $M^0L^2T^{-4}$

Q205. JEE Main 2019 (Online) 10th January Evening Slot

The diameter and height of a cylinder are measured by a meter scale to be 12.6 ± 0.1 cm and 34.2 ± 0.1 cm, respectively. What will be the value of its volume in appropriate significant figures ?

Open solution

(A) $4264.4 \pm 81.0\ cm^3$ (B) $4264 \pm 81\ cm^3$

(C) $4300 \pm 80\ cm^3$ (D) $4260 \pm 80\ cm^3$

Q206. JEE Main 2019 (Online) 10th January Morning Slot

The density of a material in SI units is 128 kg m^{-3} . In certain units in which the unit of length is 25 cm and the unit of mass is 50 g, the numerical value of density of the material is -

Open solution

(A) 40 (B) 640 (C) 16 (D) 410

Q207. JEE Main 2019 (Online) 9th January Evening Slot

Expression for time in terms of G (universal gravitational constant), h (Planck constant) and c (speed of light) is proportional to :

Open solution

(A) $\sqrt{\frac{hc^5}{G}}$ (B) $\sqrt{\frac{c^3}{Gh}}$ (C) $\sqrt{\frac{Gh}{c^5}}$ (D) $\sqrt{\frac{Gh}{c^3}}$

Q208. JEE Main 2019 (Online) 9th January Evening Slot

The pitch and the number of divisions, on the circular scale, for a given screw gauge are 0.5 mm and 100 respectively. When the screw gauge is fully tightened without any object, the zero of its circular scale lies 3 divisions below the mean line.

Open solution

The readings of the main scale and the circular scale, for a thin sheet, are 5.5 mm and 48 respectively, the thickness of this sheet is :

(A) 5.755 mm (B) 5.950 mm (C) 5.725 mm (D) 5.740 mm

Q209. JEE Main 2018 (Online) 16th April Morning Slot

The percentage errors in quantities P, Q, R and S are 0.5%, 1%, 3% and 1.5% respectively in the measurement of a physical quantity A = $\frac{P^3Q^2}{\sqrt{R}S}$.

Open solution

The maximum percentage error in the value of A will be :

(A) 6.0% (B) 7.5% (C) 8.5% (D) 6.5%

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Q210. JEE Main 2018 (Online) 16th April Morning Slot

The relative uncertainty in the period of a satellite orbiting around the earth is 10^{-2}. If the relative uncertainty in the radius of the orbit is negligible, the relative uncertainty in the mass of the earth is :

(A) 10^{-2} (B) 2×10^{-2} (C) 3×10^{-2} (D) 6×10^{-2}

Open solution

Q211. JEE Main 2018 (Offline)

The density of a material in the shape of a cube is determined by measuring three sides of the cube and its mass. If the relative errors in measuring the mass and length are respectively 1.5% and 1%, the maximum error in determining the density is:

(A) 6% (B) 2.5% (C) 3.5% (D) 4.5%

Open solution

Q212. JEE Main 2018 (Online) 15th April Evening Slot

The characteristic distance at which quantum gravitational effects are significant, the Planck length, can be determined from a suitable combination of the fundamental physical constants G, h and c.

Which of the following correctly gives the Planck length ?

(A) $G\hbar^2 c^3$ (B) $G^2 \hbar c$

(C) $G^{\frac{1}{2}}\hbar^2 c$ (D) $\left(\frac{G\hbar}{c^3}\right)^{\frac{1}{2}}$

Open solution

Q213. JEE Main 2018 (Online) 15th April Morning Slot

The relative error in the determination of the surface area of sphere is α. Then the relative error in the determination of its volume is :

(A) $\frac{3}{2}\alpha$ (B) $\frac{2}{3}\alpha$ (C) $\frac{5}{2}\alpha$ (D) α

Open solution

Q214. JEE Main 2018 (Online) 15th April Morning Slot

In a screw gauge, 5 complete rotations of the screw cause it to move a linear distance of $0.25\ cm$. There are 100 circular scale divisions. The thickness of a wire measured by this screw gauge gives a reading of 4 main scale divisions and 30 circular scale divisions. Assuming negligible zero error, the thickness of the wire is :

(A) $0.4300\ cm$ (B) $0.2150\ cm$ (C) $0.3150\ cm$ (D) $0.0430\ cm$

Open solution

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Q215. JEE Main 2017 (Online) 9th April Morning Slot

A physical quantity P is described by the relation

$P = a^{1/2} b^2 c^3 d^{-4}$

If the relative errors in the measurement of a, b, c and d respectively, are 2%, 1%, 3% and 5%, then the relative error in P will be :

(A) 8% (B) 12% (C) 32% (D) 25%

Open solution

Q216. JEE Main 2017 (Online) 8th April Morning Slot

Time (T), velocity (C) and angular momentum (h) are chosen as fundamentalquantities instead of mass, length and time. In terms of these, the dimensions of mass would be :

(A) $[M] = [T^{-1} C^{-2} h]$ (B) $[M] = [T^{-1} C^2 h]$

(C) $[M] = [T^{-1} C^{-2} h^{-1}]$ (D) $[M] = [T C^{-2} h]$

Open solution

Q217. JEE Main 2017 (Offline)

The following observations were taken for determining surface tension T of water by capillary method:

diameter of capillary, D = 1.25×10^{-2} m

rise of water, h = 1.45×10^{-2}m

Using g = 9.80 m/s^2 and the simplified relation T = $\frac{rhg}{2} \times 10^3 N/m$, the possible error in surface tension is closest to :

(A) 10 % (B) 0.15 % (C) 1.5 % (D) 2.4 %

Open solution

Q218. JEE Main 2016 (Online) 10th April Morning Slot

A thin 1 m long rod has a radius of 5 mm. A force of 50 πkN is applied at one end to determine its Young's modulus. Assume that the force is exactly known. If the least count in the measurement of all lengths is 0.01 mm, which of the following statements is **false** ?

(A) $\frac{\Delta Y}{Y}$ gets minimum contribution from the uncertainty in the length.

(B) The figure of merit is the largest for the length of the rod.

(C) The maximum value of Y that can be determined is 10^{14} N/m^2

(D) $\frac{\Delta Y}{Y}$ gets its maximum contribution from the uncertainty in strain

Open solution

Q219. JEE Main 2016 (Online) 9th April Morning Slot

In the following 'I' refers to current and other symbols have their usual meaning. Choose the option that corresponds to the dimensions of electrical conductivity :

(A) $ML^{-3} T^{-3} I^2$ (B) $M^{-1} L^3 T^3 I$ (C) $M^{-1} L^{-3} T^3 I^2$ (D) $M^{-1} L^{-3} T^3 I$

Open solution

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Q220. JEE Main 2016 (Offline)

A student measures the time period of 100 oscillations of a simple pendulum four times. The data set is 90 s, 91 s, 95 s and 92 s. If the minimum division in the measuring clock is 1 s, then the reported mean time should be :

Open solution

(A) 92 ± 1.8 s (B) 92 ± 3 s (C) 92 ± 2 s (D) 92 ± 5.0 s

Q221. JEE Main 2016 (Offline)

A screw gauge with a pitch of 0.5 mm and a circular scale with 50 divisions is used to measure the thickness of a thin sheet of Aluminium. Before starting the measurement, it is found that when the two jaws of the screw gauge are brought in contact, the 45th division coincides with the main scale line and that the zero of the main scale is barely visible. What is the thickness of the sheet if the main scale reading is 0.5 mm and the 25th division coincides with the main scale line?

Open solution

(A) 0.70 mm (B) 0.50 mm (C) 0.75 mm (D) 0.80 mm

Q222. JEE Main 2015 (Online) 11th April Morning Slot

If electronic charge e, electron mass m, speed of light in vacuum c and Planck's constant h are taken as fundamental quantities, the permeability of vacuum μ_0 can be expressed in units of :

Open solution

(A) $\left(\frac{hc}{me^2}\right)$ (B) $\left(\frac{h}{me^2}\right)$

(C) $\left(\frac{h}{ce^2}\right)$ (D) $\left(\frac{mc^2}{he^2}\right)$

Q223. JEE Main 2015 (Online) 11th April Morning Slot

A beaker contains a fluid of density ρ kg / m^3, specific heat S J/kg^{o}C and viscosity η. The beaker is filled up to height h. To estimate the rate of heat transfer per unit area ($\dot{Q}/A$) by convection when beaker is put on a hot plate, a student proposes that it should depend on η, $\left(\frac{S\Delta\theta}{h}\right)$ and $\left(\frac{1}{\rho g}\right)$ when $\Delta\theta$ (in ^{o}C) is the difference in the temperature between the bottom and top of the fluid. In that situation the correct option for $(\dot{Q}/A)$ is :

Open solution

(A) $\eta\frac{S\Delta\theta}{h}$

(B) $\eta\left(\frac{S\Delta\theta}{h}\right)\left(\frac{1}{\rho g}\right)$

(C) $\frac{S\Delta\theta}{\eta h}$

(D) $\left(\frac{S\Delta\theta}{\eta h}\right)\left(\frac{1}{\rho g}\right)$

Q224. JEE Main 2015 (Online) 10th April Morning Slot

If the capacitance of a nanocapacitor is measured in terms of a unit 'u' made by combining the electronic charge 'e', Bohr radius 'a_0 ', Planck's constant 'h' and speed of light 'c' then :

Open solution

(A) u = $\frac{e^2c}{ha_0}$ (B) u = $\frac{e^2h}{ca_0}$

(C) u = $\frac{e^2a_0}{hc}$ (D) u = $\frac{hc}{e^2a_0}$

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Q225. JEE Main 2015 (Online) 10th April Morning Slot

Open solution

Diameter of a steel ball is measured using a Vernier callipers which has divisions of 0.1 cm on its main scale (MS) and 10 divisions of its vernier scale (VS) match 9 divisions on the main scale. Three such measurements for a ball are given as :

S. No.	MS (cm)	VS divisions
1	0.5	8
2	0.5	4
3	0.5	6

If the zero error is $-$ 0.03 cm, then mean corrected diameter is :

(A) 0.56 cm **(B)** 0.59 cm **(C)** 0.53 cm **(D)** 0.52 cm

Q226. JEE Main 2015 (Offline)

Open solution

The period of oscillation of a simple pendulum is $T = 2\pi\sqrt{\frac{L}{g}}$. Measured value of L is 20.0 cm known to 1 mm accuracy and time for 100 oscillations of the pendulum is found to be 90 s using wrist watch of 1 s resolution. The accuracy in the determination of g is:

(A) 1 % **(B)** 5 % **(C)** 2 % **(D)** 3 %

Q227. JEE Main 2014 (Online) 19th April Morning Slot

Open solution

Match **List - I** (Event) with **List - II** (order of the time interval for happening of the event) and select the correct option from the options given below the lists.

List - I		List - II	
(a)	Rotation period of earth	(i)	10^5 s
(b)	Revolution period of earth	(ii)	10^7 s
(c)	Period of a light wave	(iii)	10^{-15} s
(d)	Period of a sound wave	(iv)	10^{-3} s

(A) (a)-(i), (b)-(ii), (c)-(iii), (d)-(iv) **(B)** (a)-(ii), (b)-(i), (c)-(iv), (d)-(iii)

(C) (a)-(i), (b)-(ii), (c)-(iv), (d)-(iii) **(D)** (a)-(ii), (b)-(i), (c)-(iii), (d)-(iv)

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Q228. JEE Main 2014 (Online) 12th April Morning Slot

From the following combinations of physical constant (expressed through their usual symbols) the only combination, that would have the same value in different systems of unit, is :

(A) $\frac{ch}{2\pi\in_0^2}$

(B) $\frac{e^2}{2\pi\in_0 G m_e^2}$ (m_e = mass of electron)

(C) $\frac{\mu_0 \in_0}{c^2}\frac{G}{he^2}$

(D) $\frac{2\pi\sqrt{\mu_0\in_0}}{ce^2}\frac{h}{G}$

Open solution

Q229. JEE Main 2014 (Online) 11th April Morning Slot

In terms of resistance R and time T, the dimensions of ratio $\frac{\mu}{\in}$ of the permeability μ and permittivity $\in$ is :

(A) $[RT^{-2}]$ (B) $[R^2T^{-1}]$ (C) $[R^2]$ (D) $[R^2T^2]$

Open solution

Q230. JEE Main 2014 (Online) 9th April Morning Slot

An experiment is performed to obtain the value of acceleration due to gravity g by using a simple pendulum of length L. In this experiment time for 100 oscillations is measured by using a watch of 1 second least count and the value is 90.0 seconds. The length L is measured by using a meter scale of least count 1 mm and the value is 20.0 cm. The error in the determination of g would be :

(A) 1.7 % (B) 2.7 % (C) 4.4 % (D) 2.27 %

Open solution

Q231. JEE Main 2014 (Offline)

A student measured the length of a rod and wrote it as 3.50 cm. Which instrument did he use to measure it?

(A) A screw gauge having 100 divisions in the circular scale and pitch as 1 mm.

(B) A screw gauge having 50 divisions in the circular scale and pitch as 1 mm.

(C) A meter scale.

(D) A vernier calliper where the 10 divisions in vernier scale matches with 9 divisions in main scale and main scale has 10 divisions in 1 cm.

Open solution

Q232. JEE Main 2013 (Online) 25th April Morning Slot

A metal sample carrying a current along X-axis with density J_x is subjected to a magnetic field B_z (along z-axis). The electric field E_y developed along Y-axis is directly proportional to J_x as well as B_z. The constant of proportionality has SI unit.

(A) $\frac{m^2}{A}$ (B) $\frac{m^3}{As}$ (C) $\frac{m^2}{As}$ (D) $\frac{As}{m^3}$

Open solution

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Q233. JEE Main 2013 (Online) 23th April Morning Slot

If the time period t of the oscillation of a drop of liquid of density d, radius r, vibrating under surface tension s is given by the formula $t = \sqrt{r^{2b}s^c d^{a/2}}$. It is observed that the time period is directly proportional to $\sqrt{\frac{d}{s}}$. The value of b should therefore be :

(A) $\frac{3}{4}$ (B) $\sqrt{3}$ (C) $\frac{3}{2}$ (D) $\frac{2}{3}$

Open solution

Q234. JEE Main 2013 (Online) 22th April Morning Slot

The dimensions of angular momentum, latent heat and capacitance are, respectively.

(A) $ML^2T^1A^2$, L^2T^{-2}, $M^{-1}L^{-2}T^2$
(B) ML^2T^{-2}, L^2T^2, $M^{-1}L^{-2}T^4A^2$
(C) ML^2T^{-1}, L^2T^{-2}, ML^2TA^2
(D) ML^2T^{-1}, L^2T^{-2}, $M^{-1}L^{-2}T^4A^2$

Open solution

Q235. JEE Main 2013 (Online) 9th April Morning Slot

From the following, the quantity (constructed from the basic constants of nature), that has the dimensions, as well as correct order of magnitude, vis-a-vis typical atomic size, is:

(A) $\frac{e^2}{4\pi\epsilon_0 m_e c^2}$
(B) $\frac{4\pi\epsilon_0 e^2}{m_e^2}$
(C) $\frac{me^4}{4\pi\epsilon_0\hbar^2}$
(D) $\frac{4\pi\epsilon_0 m_e c^2}{e^4}$

Open solution

Q236. JEE Main 2013 (Offline)

Let $[\varepsilon_0]$ denote the dimensional formula of the permittivity of vacuum. If M = mass, L = length, T = time and A = electric current, then:

(A) $\varepsilon_0 = \left[M^{-1}L^{-3}T^2A\right]$
(B) $\varepsilon_0 = \left[M^{-1}L^{-3}T^4A^2\right]$
(C) $\varepsilon_0 = \left[M^1L^2T^1A^2\right]$
(D) $\varepsilon_0 = \left[M^1L^2T^1A\right]$

Open solution

Q237. AIEEE 2012 (Online) 26th May

The electrical resistance R of a conductor of length l and area of cross section a is given by $R = \frac{\rho l}{a}$ where 'ρ' is the electrical resistivity. What is the dimensional formula for electrical conductivity 'σ' which is reciprocal of resistivity?

(A) $\left[M^{-1}L^{-3}T^3A^2\right]$
(B) $\left[ML^{-3}T^{-3}A^2\right]$
(C) $\left[ML^3T^{-3}A^{-2}\right]$
(D) $\left[M^{-2}L^3T^2A^{-1}\right]$

Open solution

Q238. AIEEE 2012 (Online) 19th May

The amount of heat produced in an electric circuit depends upon the current (I), resistance (R) and time (t). If the error made in the measurements of the above quantities are 2%, 1% and 1% respectively then the maximum possible error in the total heat produced will be

(A) 1% (B) 2% (C) 6% (D) 3%

Open solution

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Q239. AIEEE 2012 (Online) 19th May

N divisions on the main scale of a vernier calliper coincide with $(N+1)$ divisions of the vernier scale. If each division of main scale is 'a' units, then the least count of the instrument is

(A) a (B) $\frac{a}{N}$ (C) $\frac{N}{N+1} \times a$ (D) $\frac{a}{N+1}$

Open solution

Q240. AIEEE 2012 (Online) 12th May

A student measured the diameter of a wire using a screw gauge with the least count $0.001\ \text{cm}$ and listed the measurements. The measured value should be recorded as

(A) $5.3200\ \text{cm}$ (B) $5.3\ \text{cm}$ (C) $5.32\ \text{cm}$ (D) $5.320\ \text{cm}$

Open solution

Q241. AIEEE 2012 (Online) 7th May

Given that K = energy, V = velocity, T = time. If they are chosen as the fundamental units, then what is dimensional formula for surface tension?

(A) $\left[KV^{-2}T^{-2}\right]$ (B) $\left[K^2V^2T^{-2}\right]$ (C) $\left[K^2V^{-2}T^{-2}\right]$ (D) $\left[KV^2T^2\right]$

Open solution

Q242. AIEEE 2012

Resistance of a given wire is obtained by measuring the current flowing in it and the voltage difference applied across it. If the percentage errors in the measurement of the current and the voltage difference are 3% each, then error in the value of resistance of the wire is

(A) 6 % (B) zero (C) 1 % (D) 3 %

Open solution

Q243. AIEEE 2012

A spectrometer gives the following reading when used to measure the angle of a prism.
Main scale reading: 58.5 degree
Vernier scale reading : 09 divisions
Given that 1 division on main scale corresponds to 0.5 degree. Total divisions on the vernier scale is 30 and match with 29 divisions of the main scale. The angle of the prism from the above data is

(A) 58.59 degree (B) 58.77 degree (C) 58.65 degree (D) 59 degree

Open solution

Q244. AIEEE 2011

A screw gauge gives the following reading when used to measure the diameter of a wire.
Main scale reading : 0 mm
Circular scale reading : 52 divisions
Given that 1 mm on main scale corresponds to 100 divisions of the circular scale.
The diameter of wire from the above data is:

(A) 0.052 cm (B) 0.026 cm (C) 0.005 cm (D) 0.52 cm

Open solution

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Q245. AIEEE 2010

The respective number of significant figures for the numbers 23.023, 0.0003 and 2.1×10^{-3} are

(A) 5, 1, 2 (B) 5, 1, 5 (C) 5, 5, 2 (D) 4, 4, 2

Open solution

Q246. AIEEE 2009

In an experiment the angles are required to be measured using an instrument, 29 divisions of the main scale exactly coincide with the 30 divisions of the vernier scale. If the smallest division of the main scale is half-a degree($=0.5^{\circ}$), then the least count of the instrument is:

(A) one minute (B) half minute (C) one degree (D) half degree

Open solution

Q247. AIEEE 2008

The dimension of magnetic field in M, L, T and C (coulomb) is given as

(A) $MLT^{-1}C^{-1}$ (B) $MT^{2}C^{-2}$ (C) $MT^{-1}C^{-1}$ (D) $MT^{-2}C^{-1}$

Open solution

Q248. AIEEE 2008

A body of mass m = 3.513 kg is moving along the x-axis with a speed of 5.00 ms^{-1}. The magnitude of its momentum is recorded as

(A) 17.6 kg ms^{-1} (B) 17.565 kg ms^{-1} (C) 17.56 kg ms^{-1} (D) 17.57 kg ms^{-1}

Open solution

Q249. AIEEE 2008

Two full turns of the circular scale of a screw gauge cover a distance of 1 mm on its main scale. The total number of divisions on the circular scale is 50. Further, it is found that the screw gauge has a zero error of − 0.03 mm. While measuring the diameter of a thin wire, a student notes the main scale reading of 3 mm and the number of circular scale divisions in line with the main scale as 35. The diameter of the wire is

(A) 3.32 mm (B) 3.73 mm (C) 3.67 mm (D) 3.38 mm

Open solution

Q250. AIEEE 2008

An experiment is performed to find the refractive index of glass using a travelling microscope. In this experiment distances are measured by

(A) a vernier scale provided on the microscope

(B) a standard laboratory scale

(C) a meter scale provided on the microscope

(D) a screw gauge provided on the microscope

Open solution

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Q251. AIEEE 2006

Which of the following units denotes the dimension $\frac{ML^2}{Q^2}$, where Q denotes the electric charge?

(A) Wb/m^2 (B) Henry (H) (C) H/m^2 (D) Weber (Wb)

Open solution

Q252. AIEEE 2005

Out of the following pair, which one does NOT have identical dimensions is

(A) angular momentum and Planck's constant (B) impulse and momentum

(C) moment of inertia and moment of a force (D) work and torque

Open solution

Q253. AIEEE 2004

Which one of the following represents the correct dimensions of the coefficient of viscosity?

(A) $ML^{-1}T^{-1}$ (B) MLT^{-1} (C) $ML^{-1}T^{-2}$ (D) $ML^{-2}T^{-2}$

Open solution

Q254. AIEEE 2003

Dimensions of $\frac{1}{\mu_0\varepsilon_0}$, where symbols have their usual meaning, are

(A) $[L^{-1}T]$ (B) $[L^{-2}T^2]$ (C) $[L^2T^{-2}]$ (D) $[LT^{-1}]$

Open solution

Q255. AIEEE 2003

The physical quantities not having same dimensions are

(A) torque and work (B) momentum and Planck's constant

(C) stress and Young's modulus (D) speed and $(\mu_0\varepsilon_0)^{-1/2}$

Open solution

Q256. AIEEE 2002

Identify the pair whose dimensions are equal

(A) torque and work (B) stress and energy

(C) force and stress (D) force and work

Open solution

Numerical 24 questions

Q257. JEE Main 2026 (Online) 23rd January Evening Shift

A ball of radius r and density ρ dropped through a viscous liquid of density σ and viscosity η attains its terminal velocity at time t, given by $t = A\rho^a r^b \eta^c \sigma^d$, where A is a constant and a, b, c and d are integers. The value of $\frac{b+c}{a+d}$ is ____ .

Open solution

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Q258. JEE Main 2025 (Online) 3rd April Evening Shift

A physical quantity C is related to four other quantities $\mathrm{p}, \mathrm{q}, \mathrm{r}$ and s as follows

$C = \frac{pq^2}{r^3\sqrt{s}}$

The percentage errors in the measurement of $\mathrm{p}, \mathrm{q}, \mathrm{r}$ and s are $1\%, 2\%, 3\%$ and 2%, respectively. The percentage error in the measurement of C will be_____%

Open solution

Q259. JEE Main 2025 (Online) 29th January Evening Shift

A physical quantity Q is related to four observables a, b, c, d as follows :

$Q = \frac{ab^4}{cd}$

where, $\mathrm{a} = (60 \pm 3)\mathrm{Pa}; \mathrm{b} = (20 \pm 0.1)\mathrm{m}; \mathrm{c} = (40 \pm 0.2)\mathrm{Nsm}^{-2}$ and $\mathrm{d} = (50 \pm 0.1)\mathrm{m}$, then the percentage error in Q is $\frac{x}{1000}$, where $x =$ _____ .

Open solution

Q260. JEE Main 2025 (Online) 28th January Morning Shift

In a measurement, it is asked to find modulus of elasticity per unit torque applied on the system. The measured quantity has dimension of $\left[M^a L^b T^c\right]$. If $b = -3$, the value of c is _____.

Open solution

Q261. JEE Main 2025 (Online) 28th January Morning Shift

A tiny metallic rectangular sheet has length and breadth of 5 mm and 2.5 mm , respectively. Using a specially designed screw gauge which has pitch of 0.75 mm and 15 divisions in the circular scale, you are asked to find the area of the sheet. In this measurement, the maximum fractional error will be $\frac{x}{100}$ where x is _____ .

Open solution

Q262. JEE Main 2025 (Online) 24th January Morning Shift

The least count of a screw gauge is 0.01 mm . If the pitch is increased by 75% and number of divisions on the circular scale is reduced by 50%, the new least count will be _____ $\times 10^{-3}$ mm

Open solution

Q263. JEE Main 2023 (Online) 30th January Morning Shift

In an experiment for estimating the value of focal length of converging mirror, image of an object placed at $40\ \mathrm{cm}$ from the pole of the mirror is formed at distance $120\ \mathrm{cm}$ from the pole of the mirror. These distances are measured with a modified scale in which there are 20 small divisions in $1\ \mathrm{cm}$. The value of error in measurement of focal length of the mirror is $\frac{1}{\mathrm{K}}\ \mathrm{cm}$. The value of K is _____.

Open solution

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Q264. JEE Main 2023 (Online) 30th January Morning Shift

In a screw gauge, there are 100 divisions on the circular scale and the main scale moves by $0.5\ \text{mm}$ on a complete rotation of the circular scale. The zero of circular scale lies 6 divisions below the line of graduation when two studs are brought in contact with each other. When a wire is placed between the studs, 4 linear scale divisions are clearly visible while 46th division the circular scale coincide with the reference line. The diameter of the wire is _____ $\times 10^{-2}\ \text{mm}$.

Open solution

Q265. JEE Main 2022 (Online) 28th July Evening Shift

In an experiment to find acceleration due to gravity (g) using simple pendulum, time period of $0.5\ \text{s}$ is measured from time of 100 oscillation with a watch of $1\ \text{s}$ resolution. If measured value of length is $10\ \text{cm}$ known to $1\ \text{mm}$ accuracy, The accuracy in the determination of g is found to be $x\%$. The value of x is _____.

Open solution

Q266. JEE Main 2022 (Online) 27th July Morning Shift

The one division of main scale of Vernier callipers reads $1\ \text{mm}$ and 10 divisions of Vernier scale is equal to the 9 divisions on main scale. When the two jaws of the instrument touch each other, the zero of the Vernier lies to the right of zero of the main scale and its fourth division coincides with a main scale division. When a spherical bob is tightly placed between the two jaws, the zero of the Vernier scale lies in between $4.1\ \text{cm}$ and $4.2\ \text{cm}$ and 6th Vernier division coincides scale division. The diameter of the bob will be _____ $\times\ 10^{-2}$ cm.

Open solution

Q267. JEE Main 2022 (Online) 26th July Morning Shift

In an experiment to determine the Young's modulus of wire of a length exactly $1\ \text{m}$, the extension in the length of the wire is measured as $0.4\ \text{mm}$ with an uncertainty of $\pm 0.02\ \text{mm}$ when a load of $1\ \text{kg}$ is applied. The diameter of the wire is measured as $0.4\ \text{mm}$ with an uncertainty of $\pm 0.01\ \text{mm}$. The error in the measurement of Young's modulus (ΔY) is found to be $x \times 10^{10}\ \text{Nm}^{-2}$. The value of x is _____. (take $\text{g} = 10\ \text{ms}^{-2}$)

Open solution

Q268. JEE Main 2022 (Online) 29th June Evening Shift

The Vernier constant of Vernier callipers is 0.1 mm and it has zero error of (-0.05) cm. While measuring diameter of a sphere, the main scale reading is 1.7 cm and coinciding vernier division is 5. The corrected diameter will be _____ $\times\ 10^{-2}$ cm.

Open solution

Q269. JEE Main 2022 (Online) 28th June Evening Shift

A student in the laboratory measures thickness of a wire using screw gauge. The readings are 1.22 mm, 1.23 mm, 1.19 mm and 1.20 mm. The percentage error is $\frac{x}{121}\%$. The value of x is _____.

Open solution

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Q270. JEE Main 2022 (Online) 26th June Evening Shift

A travelling microscope is used to determine the refractive index of a glass slab. If 40 divisions are there in 1 cm on main scale and 50 Vernier scale divisions are equal to 49 main scale divisions, then least count of the travelling microscope is _____ $\times 10^{-6}$ m.

Open solution

Q271. JEE Main 2022 (Online) 26th June Morning Shift

In a vernier callipers, each cm on the main scale is divided into 20 equal parts. If tenth vernier scale division coincides with nineth main scale division. Then the value of vernier constant will be _____ $\times 10^{-2}$ mm.

Open solution

Q272. JEE Main 2022 (Online) 25th June Evening Shift

For $z = a^2x^3y^{\frac{1}{2}}$, where 'a' is a constant. If percentage error in measurement of 'x' and 'y' are 4% and 12% respectively, then the percentage error for 'z' will be _____%.

Open solution

Q273. JEE Main 2021 (Online) 31st August Evening Shift

The diameter of a spherical bob is measured using a vernier callipers. 9 divisions of the main scale, in the vernier callipers, are equal to 10 divisions of vernier scale. One main scale division is 1 mm. The main scale reading is 10 mm and 8th division of vernier scale was found to coincide exactly with one of the main scale division. If the given vernier callipers has positive zero error of 0.04 cm, then the radius of the bob is _____ $\times 10^{-2}$ cm.

Open solution

Q274. JEE Main 2021 (Online) 26th August Evening Shift

The acceleration due to gravity is found upto an accuracy of 4% on a planet. The energy supplied to a simple pendulum of known mass 'm' to undertake oscillations of time period T is being estimated. If time period is measured to an accuracy of 3%, the accuracy to which E is known is .....%

Open solution

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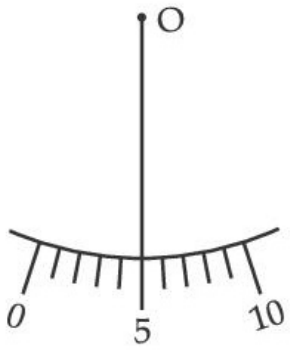
Q275. JEE Main 2021 (Online) 25th July Morning Shift

Open solution

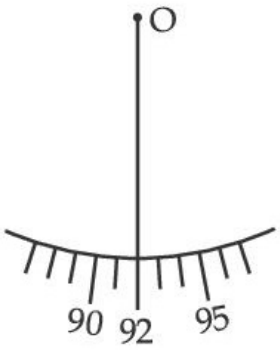
Student A and student B used two screw gauges of equal pitch and 100 equal circular divisions to measure the radius of a given wire. The actual value of the radius of the wire is 0.322 cm. The absolute value of the difference between the final circular scale readings observed by the students A and B is _____.

[Figure shows position of reference 'O' when jaws of screw gauge are closed]

Given pitch = 0.1 cm.



Screw gauge
(A)



Screw gauge
(B)

Q276. JEE Main 2021 (Online) 22th July Evening Shift

Open solution

Three students S_1, S_2 and S_3 perform an experiment for determining the acceleration due to gravity (g) using a simple pendulum. They use different lengths of pendulum and record time for different number of oscillations. The observations are as shown in the table.

Student No.	Length of Pendulum (cm)	No. of oscillations (n)	Total time for n oscillations	Time period (s)
1	64.0	8	128.0	16.0
2	64.0	4	64.0	16.0
3	20.0	4	36.0	9.0

(Least count of length = 0.1 cm and Least count for time = 0.1 s)

If E_1, E_2 and E_3 are the percentage errors in 'g' for students 1, 2 and 3 respectively, then the minimum percentage error is obtained by student no. _____.

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Q277. JEE Main 2021 (Online) 18th March Evening Shift

The radius of a sphere is measured to be (7.50 $\pm$ 0.85) cm. Suppose the percentage error in its volume is x.

The value of x, to the nearest integer, is _____.

Open solution

Q278. JEE Main 2021 (Online) 17th March Evening Shift

Suppose you have taken a dilute solution of oleic acid in such a way that its concentration becomes 0.01 cm^3 of oleic acid per cm^3 of the solution. Then you make a thin film of this solution (monomolecular thickness) of area 4 cm^2 by considering 100 spherical drops of radius $\left(\frac{3}{40\pi}\right)^{\frac{1}{3}} \times 10^{-3}$ cm. Then the thickness of oleic acid layer will be x $\times$ 10^{-14} m. Where x is _____.

Open solution

Q279. JEE Main 2021 (Online) 16th March Morning Shift

The resistance R = $\frac{V}{I}$, where V = (50 $\pm$ 2)V and I = (20 $\pm$ 0.2)A. The percentage error in R is 'x'%. The value of 'x' to the nearest integer is _____.

Open solution

Q280. JEE Main 2020 (Online) 6th September Morning Slot

The density of a solid metal sphere is determined by measuring its mass and its diameter. The maximum error in the density of the sphere is $\left(\frac{x}{100}\right)$ %. If the relative errors in measuring the mass and the diameter are 6.0% and 1.5% respectively, the value of x is_____.

Open solution

MCQ (More than One Correct Answer) 1 questions

Q281. JEE Main 2016 (Online) 10th April Morning Slot

A, B, C and D are four different physical quantities having different dimensions. None of them is dimensionless. But we know that the equation AD = C ln (BD) holds true. Then which of the combination is **not** a meaningful quantity ?

Open solution

(A) $A^2 - B^2C^2$

(B) $\frac{(A-C)}{D}$

(C) $\frac{A}{B} - C$

(D) $\frac{C}{BD} - \frac{AD^2}{C}$

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Answer Key

MCQ (Single Correct Answer)

1. **B**	2. **A**	3. **C**	4. **C**	5. **B**
6. **D**	7. **B**	8. **D**	9. **C**	10. **A**
11. **A**	12. **D**	13. **B**	14. **A**	15. **B**
16. **B**	17. **D**	18. **C**	19. **B**	20. **A**
21. **D**	22. **A**	23. **D**	24. **C**	25. **A**
26. **C**	27. **C**	28. **D**	29. **C**	30. **A**
31. **C**	32. **D**	33. **C**	34. **C**	35. **D**
36. **B**	37. **A**	38. **D**	39. **A**	40. **D**
41. **A**	42. **B**	43. **D**	44. **C**	45. **C**
46. **D**	47. **C**	48. **B**	49. **C**	50. **C**
51. **D**	52. **A**	53. **A**	54. **C**	55. **A**
56. **C**	57. **C**	58. **B**	59. **B**	60. **C**
61. **C**	62. **B**	63. **A**	64. **C**	65. **D**
66. **B**	67. **A**	68. **B**	69. **D**	70. **D**
71. **D**	72. **A**	73. **D**	74. **B**	75. **D**
76. **B**	77. **D**	78. **B**	79. **C**	80. **C**
81. **D**	82. **A**	83. **A**	84. **C**	85. **D**
86. **B**	87. **A**	88. **D**	89. **B**	90. **A**
91. **C**	92. **D**	93. **B**	94. **D**	95. **B**
96. **A**	97. **B**	98. **A**	99. **D**	100. **B**
101. **A**	102. **C**	103. **D**	104. **D**	105. **A**
106. **B**	107. **B**	108. **C**	109. **C**	110. **A**
111. **C**	112. **C**	113. **D**	114. **B**	115. **A**
116. **C**	117. **A**	118. **D**	119. **A**	120. **B**
121. **B**	122. **B**	123. **B**	124. **D**	125. **C**
126. **C**	127. **D**	128. **C**	129. **D**	130. **B**
131. **C**	132. **B**	133. **D**	134. **A**	135. **A**
136. **A**	137. **C**	138. **A**	139. **A**	140. **A**
141. **C**	142. **C**	143. **C**	144. **A**	145. **D**
146. **B**	147. **C**	148. **B**	149. **A**	150. **A**

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151. **B**	152. **A**	153. **C**	154. **A**	155. **D**
156. **C**	157. **C**	158. **D**	159. **A**	160. **C**
161. **A**	162. **B**	163. **D**	164. **D**	165. **A**
166. **B**	167. **B**	168. **C**	169. **A**	170. **C**
171. **A**	172. **C**	173. **A**	174. **C**	175. **C**
176. **C**	177. **A**	178. **C**	179. **B**	180. **B**
181. **C**	182. **A**	183. **B**	184. **B**	185. **C**
186. **C**	187. **A**	188. **D**	189. **A**	190. **D**
191. **A**	192. **B**	193. **C**	194. **C**	195. **A**
196. **A**	197. **C**	198. **D**	199. **A**	200. **B**
201. **A**	202. **C**	203. **C**	204. **B**	205. **D**
206. **A**	207. **C**	208. **C**	209. **D**	210. **B**
211. **D**	212. **D**	213. **A**	214. **B**	215. **C**
216. **A**	217. **C**	218. **C**	219. **C**	220. **C**
221. **D**	222. **C**	223. **A**	224. **C**	225. **B**
226. **D**	227. **A**	228. **B**	229. **C**	230. **B**
231. **D**	232. **B**	233. **C**	234. **D**	235. **A**
236. **B**	237. **A**	238. **C**	239. **D**	240. **D**
241. **A**	242. **A**	243. **C**	244. **A**	245. **A**
246. **A**	247. **C**	248. **A**	249. **D**	250. **A**
251. **B**	252. **C**	253. **A**	254. **C**	255. **B**
256. **A**				

Numerical

257. **1**	258. **15**	259. **7700**	260. **0**	261. **3**
262. **35**	263. **32**	264. **220**	265. **5**	266. **412**
267. **2**	268. **180**	269. **150**	270. **5**	271. **5**
272. **18**	273. **52**	274. **14**	275. **13**	276. **1**
277. **34**	278. **25**	279. **5**	280. **1050**	

MCQ (More than One Correct Answer)

281. **B, D**

For numerical answers given as a range, any value within the range is correct. Full solutions: scan the QR code next to each question, or visit questions.examside.com.

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